Class - 13

D.C. Machine

12 Feb 2026

$e = Blxv$

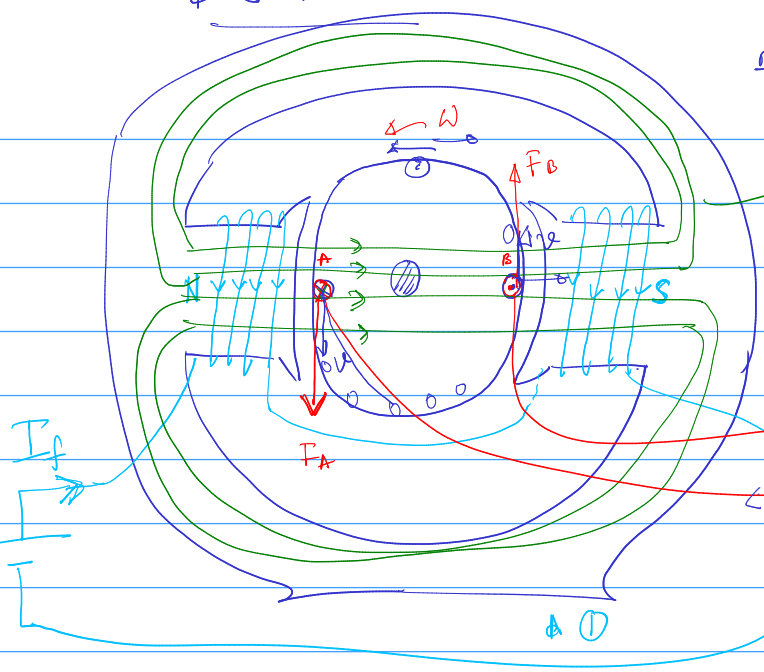
(b)

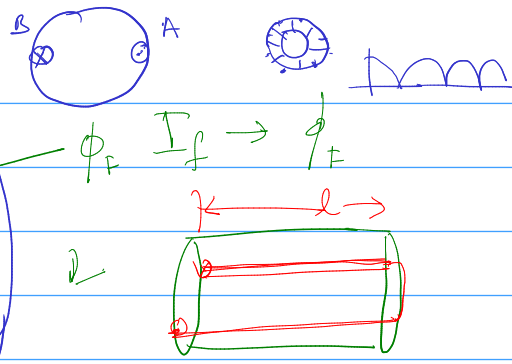
$F_A = I l \times B$

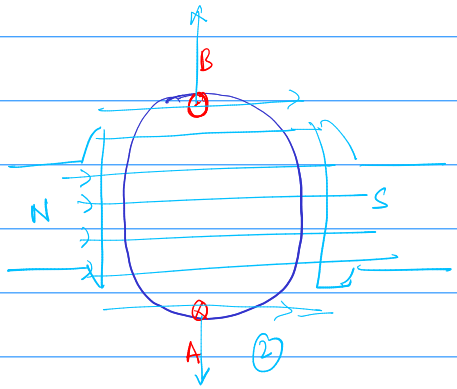
$= (l \cdot I_A \cdot B)$

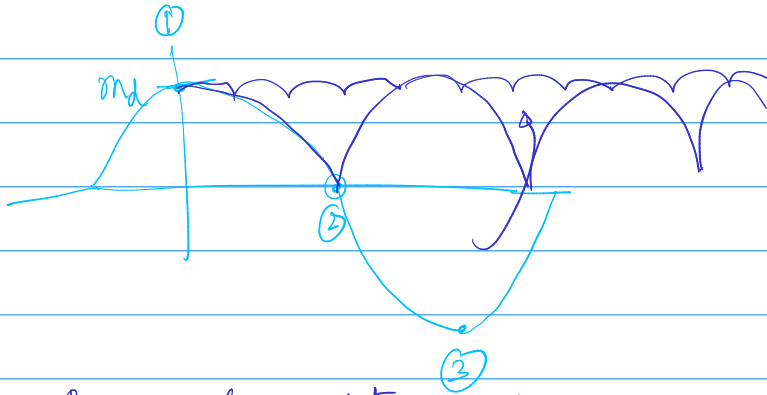
$F_B = (l \cdot I_A \cdot B)$

$F_A = F_B$

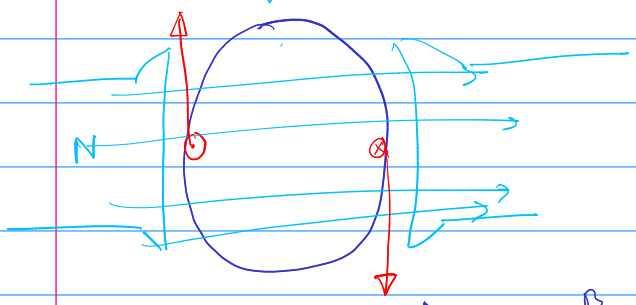


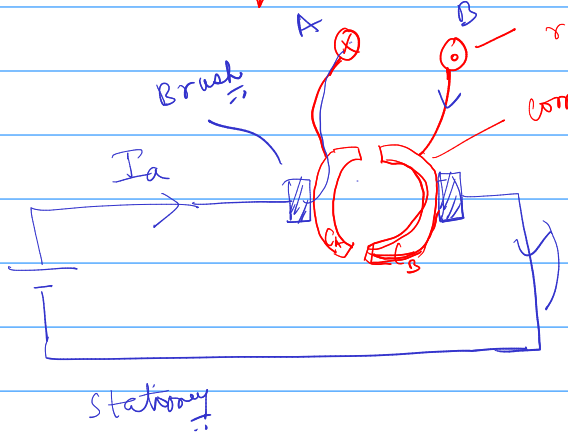


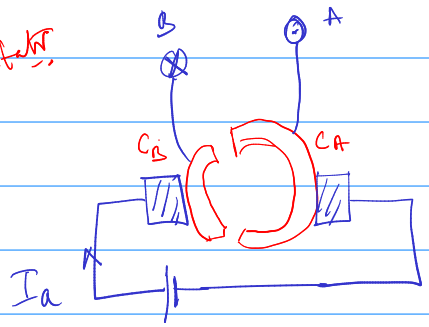




Brush & Commutator segment

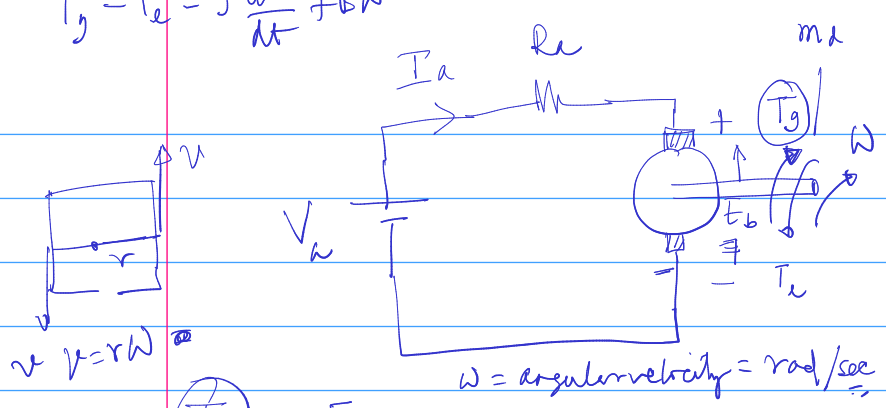






$T_g - T_e = J\frac{d\omega}{dt} + B\omega$

(J) = inertia of the motor, B = friction coeffic



m_d / T_e

$T_g = mA$

T_g = generated torque

m_d = developed torque

T_e = load torque

ω = angular velocity = rad/sec

ω = angular velocity.

K_t = Torque constant

T_g, E_b

$T_g \propto I_f$; $T_g \propto I_a$

$T_g = K_t I_f \cdot I_a$

$E_b \propto I_f$; $E_b \propto \omega$

$E_b \propto I_f \cdot \omega$

$E_b = K_b \cdot I_f \cdot \omega$

$V_f = R_f I_f + L_f \frac{dI_f}{dt}$ → (1)

$V_a = R_a I_a + E_b - L_a \frac{dI_a}{dt}$ → (2)

$E_b = K_b I_f \omega$ → (3)

$T_g = K_t \cdot I_f \cdot I_a$ → (4)

$T_g = T_e + J\frac{d\omega}{dt} + B\omega$ → (5)

J = Moment of inertia

B = frictional coff.

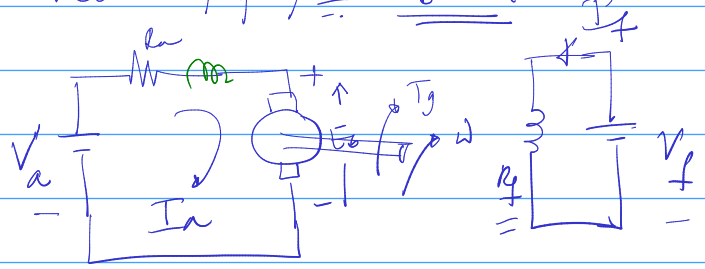
Class - 14

13/02/2026

$E_b = K_b \cdot I_f \cdot \omega$

$T_g = K_t \cdot I_f \cdot I_a$

relationship b/wn K_b & K_t



Power input $V_a I_a$ $\rightarrow$ $I_a^2 R_a$ + ($E_b \cdot I_a$) $\Rightarrow$ ($T_g \omega$)

electrical power getting converted to mechanical power

developed mechanical power

$E_b I_a = T_g \cdot \omega$

$K_b I_f \cdot \phi \cdot I_a = K_t I_f \cdot I_a \cdot \omega$

E_b $K_\phi = K_t$

$V = I_a R_a + K I_f \cdot \omega \rightarrow (1)$

$V_f = R_f I_f$ (3)

$T_g = K I_f \cdot I_a \rightarrow (2)$

$T_g = T_L + J\frac{d\omega}{dt} + B\omega \rightarrow (4)$

$I_f \propto \phi$

$V = I_a R_a + K \phi \omega \rightarrow (1)$

$T_g = K \phi I_a \rightarrow (2)$

Electrical Motors Torque vs speed characteristics

(T_g) (ω)

$V_a = R_a I_a + K\phi\omega$

$T_g = K\phi I_a$

$V_a = R_a \frac{T_g}{(K\phi)} + K\phi\omega$

$\frac{R_a}{(K\phi)} T_g = V_a - K\phi\omega$

$T_g = \left(\frac{K\phi}{R_a}\right) V_a - \frac{(K\phi)^2}{R_a} \omega \rightarrow (3)$

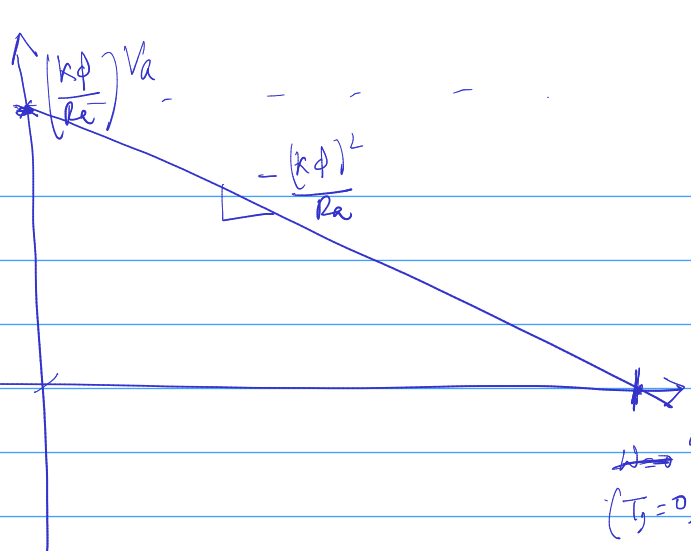
$y = c \cdot + mx$

$T_g = \left(\frac{k\phi}{R_a}\right) V_a - \frac{(k\phi)^2}{R_a}(\omega)$

$T_g = 0$

$\omega \frac{(k\phi)^2}{R_a} = \left(\frac{k\phi}{R_a}\right) V_a$

$\omega = \left(\frac{V_a}{k\phi}\right)$



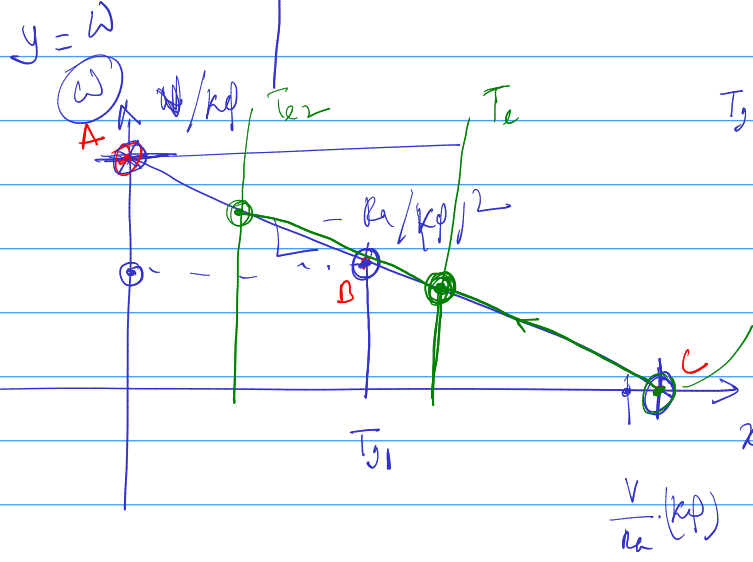
$T_g = (k\phi) \cdot I_a$

$I_a = 0, T_g = 0$

$I_a = \frac{V}{R_a}$

$x = (T_g)$ $I_a = \frac{V - (k\phi)\omega}{R_a}$

I_a (start) $\ll I_a$ (start)



$T_g > T_e$

$T_g = T_e + J\frac{d\omega}{dt} + B\omega$

$I_a = \frac{V - (k\phi)\omega}{R_a}$

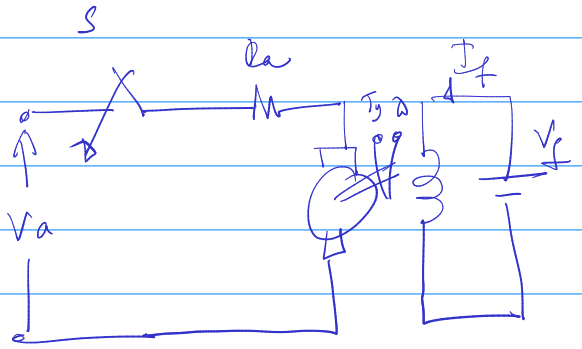
$V = I_a R_a + k\phi\omega$

$T_g = k\phi I_a$

$\omega = \frac{V}{(k\phi)} - \frac{I_a R_a}{(k\phi)}$

$\omega = \frac{V}{(k\phi)} - \frac{T_g R_a}{(k\phi)^2}$

$y = c + mx$



$I_a = (V_a/R_a)$ $E_b = (k\phi) \cdot \omega$

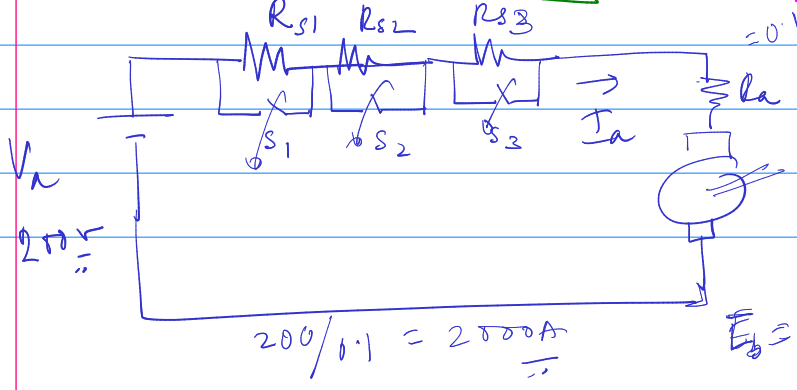
When $\omega = 0$

$T_g \frac{R_a}{(k\phi)^2} = \frac{V}{k\phi} \Rightarrow T_g = \left(\frac{V(k\phi)}{R_a}\right)$

$R_a = 0.1\Omega$

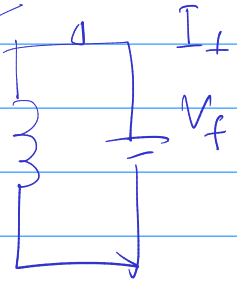
10A

Armature resistance starting



Separately excited D.C machine $I_{a(start)} = \frac{200}{0.1}$

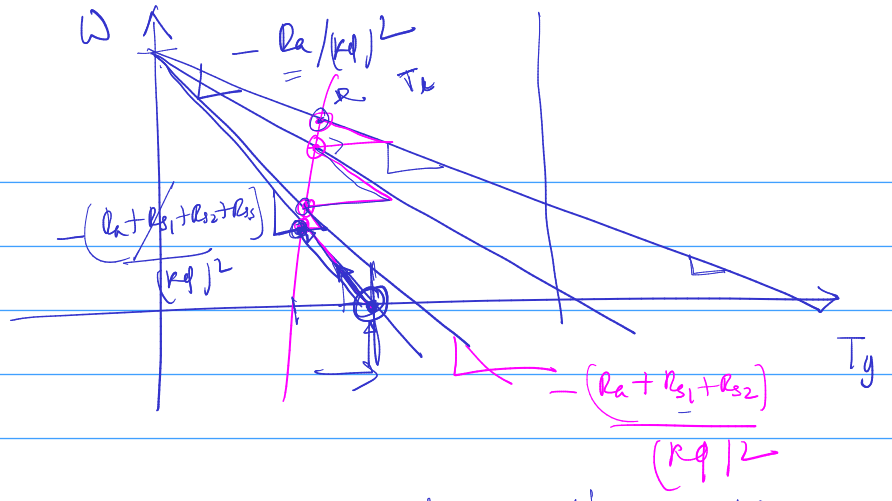
$= 2000A$



$R_{s1} + R_{s2} + R_{s3} = 20\Omega$

$I_{a(start)} = \frac{200}{20.1}$

$\simeq 10A$



1) Starting — resistive starting — disadv high loss
2) Speed control
3) Stop

speed control of DC machine with resistance

$V_a = R_a I_a + K\phi\omega$

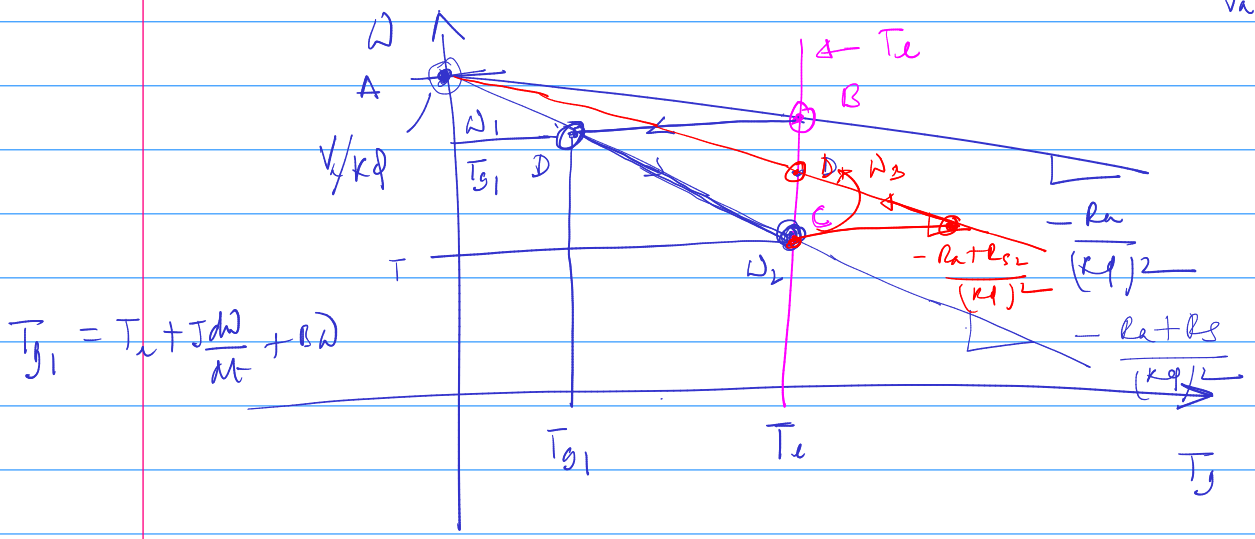
no load

$T = K\phi I_a$

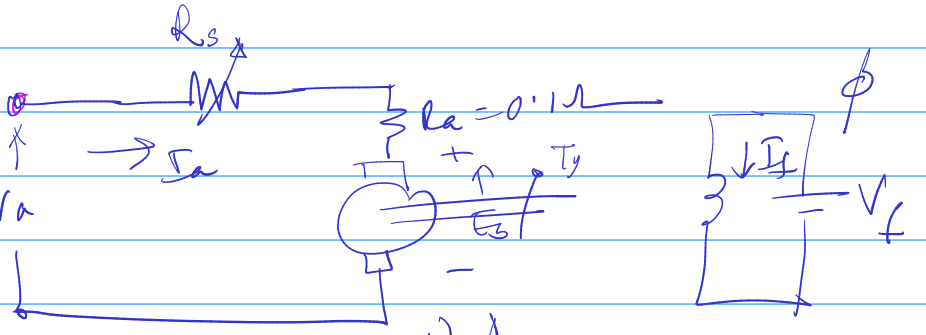
$I_a = 0$

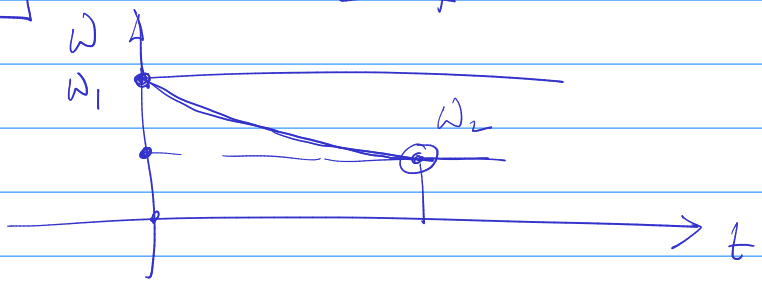
$T = 0$

$\omega = V_a/K\phi$



$T_{g1} = T_L + J\frac{d\omega}{dt} + B\omega$





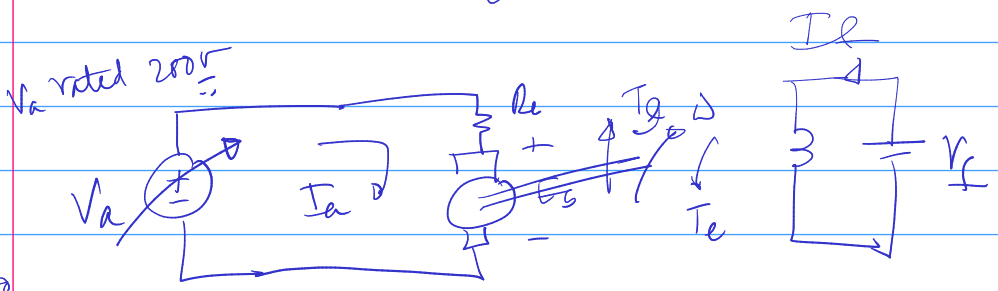
17th class

19/02/2026

$$\omega = \frac{V_a}{(K\phi)} - \frac{I_a R_a}{(K\phi)^2} \rightarrow (1)$$

change with V

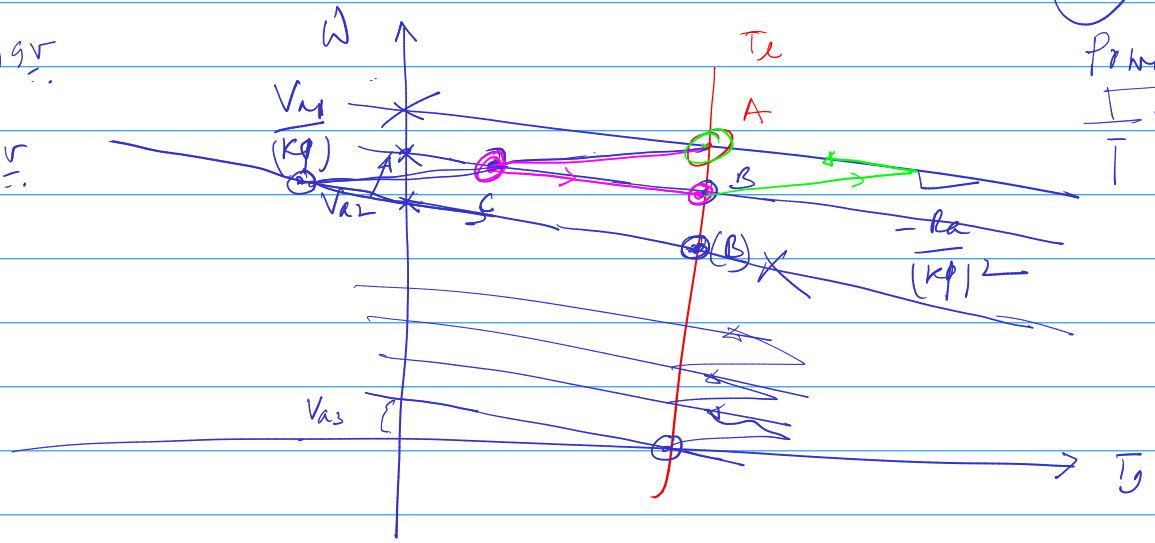
change with $(K\phi)$



$V_a = 200$ V

$I_f = 1.95$ A

$V_a = 180$ V



V_a

Power electronics

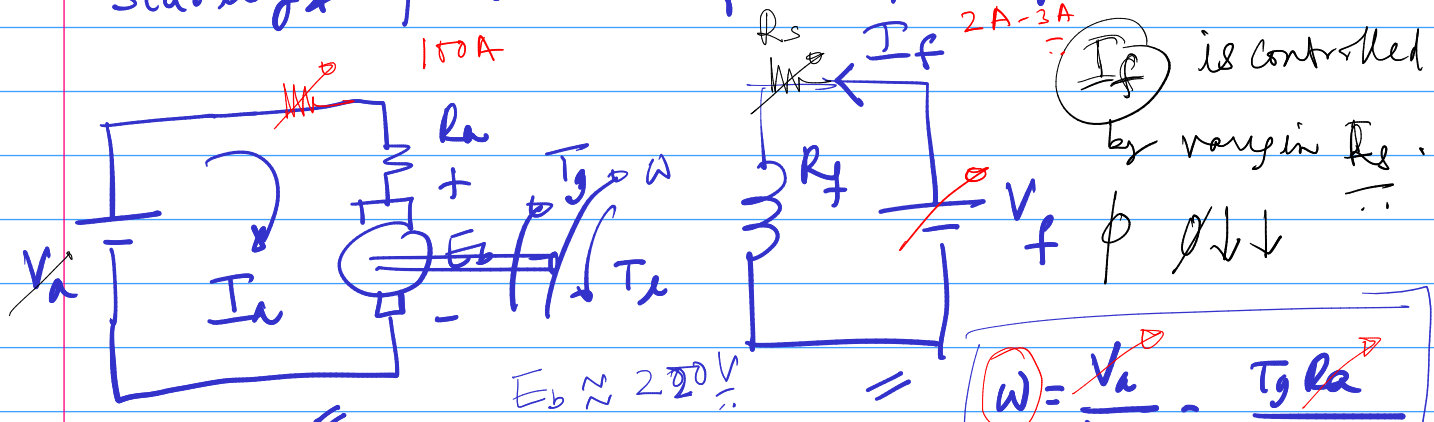
V_a V_b

18th class

20/02/2026

Quiz1 syllabus: Transformer & Magnetic circuit

Starting & speed control of a separately excited DC Machine



I_f is controlled by varying R_s.

$$\omega = \frac{V_a}{(K\phi)} - \frac{T_g R_a}{(K\phi)^2}$$

1) Armature resistance control.

2) Armature voltage control (R_a is fixed) we control the terminal voltage of the m/c

3) field weakening of a DC mtr

$$T = (K\phi) I_a \uparrow$$

In field weakening generated torque is less that the rated torque

Name plate

220 V, 100 KW, (1000 rpm) separately excited DC. mtr

$$T_g = \frac{100 \times 1000}{100}$$

$$= 1000 \text{ (N-m)}$$

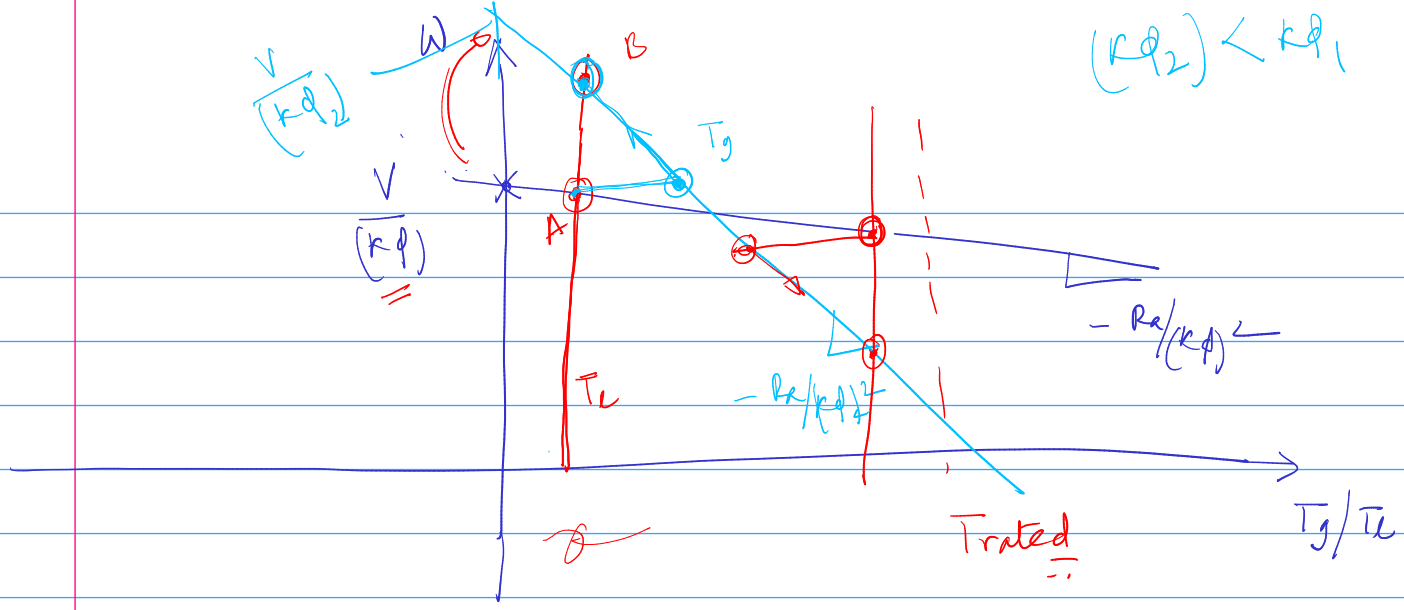
1000 rpm → 1500 rpm

1000 rpm = rad/sec

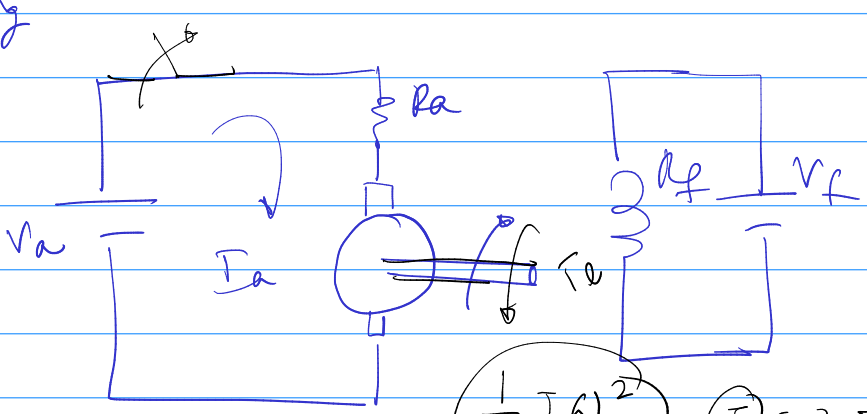
$$\frac{1000 * 2\pi}{3\phi} \text{ rad/sec}$$

$$= 33.33 * \pi \text{ rad/sec}$$

$$\simeq 100 \text{ rad/sec}$$



Braking



$\frac{1}{2}J\omega^2$ (J) = moment of inertia

$T_g = 0$

$B\omega$

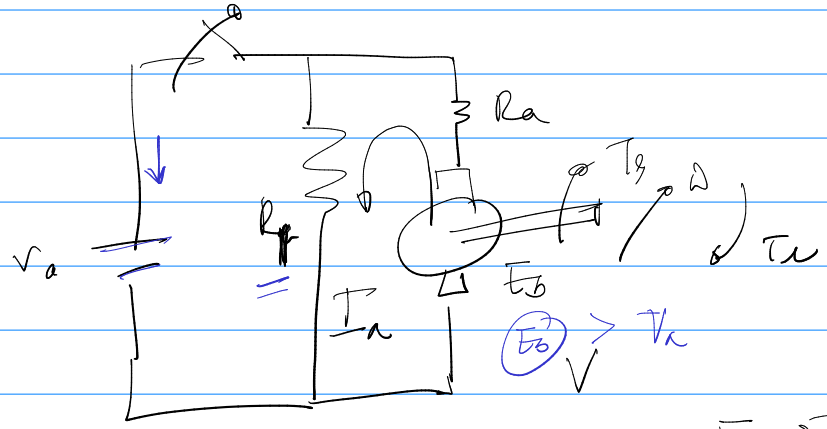
$B = 0$

J

Regenerative Braking

, lossy Braking by connecting an external resistor with the supply.

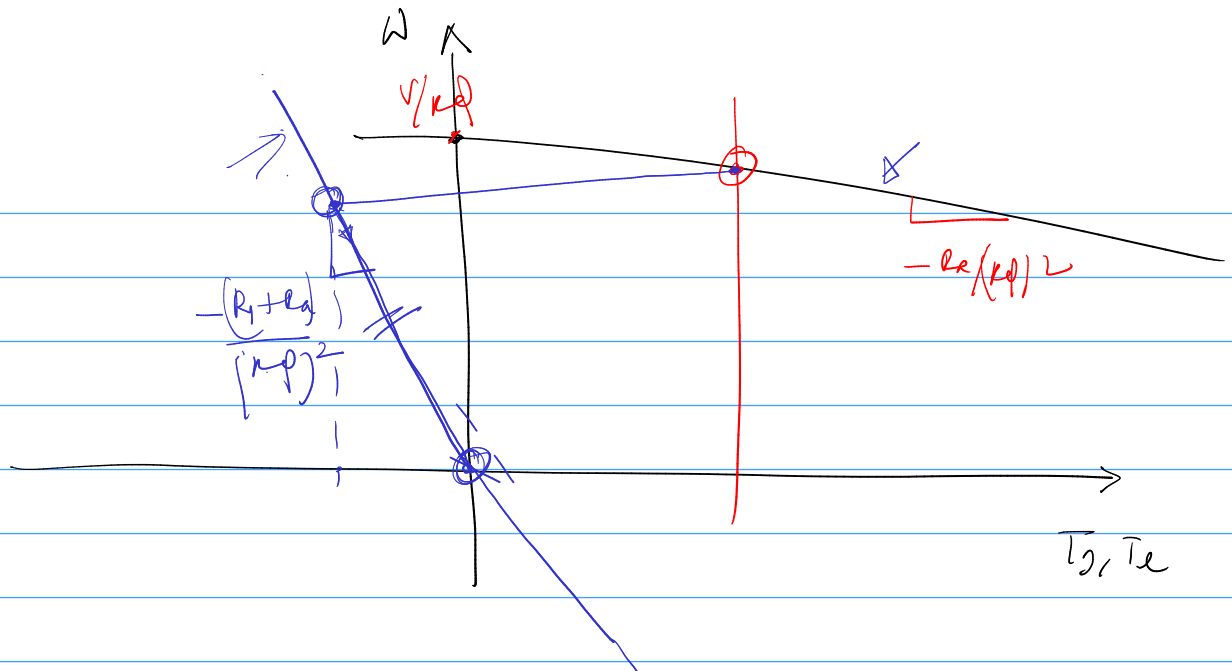
Resistive Braking



$T_g = (k\phi I_a)$

$E_b \rightarrow I_a$ will become negative

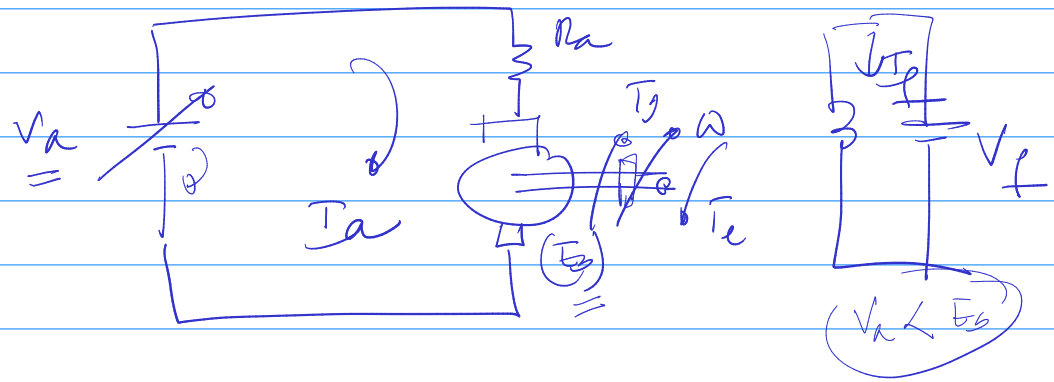
$$I_a = \frac{E_b}{(R_a + R_f)}$$

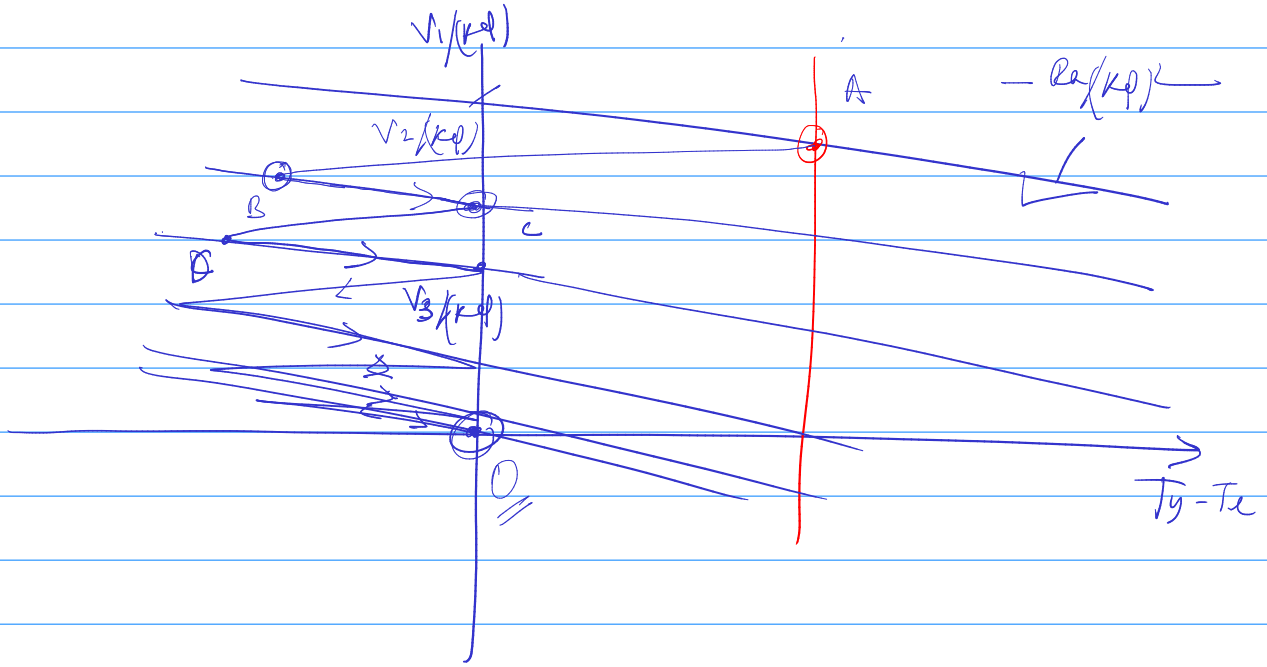


Regenerative braking: We recover the energy from the machine into the battery itself.

(i) $V_a < E_b$, which will cause I_a in -ve direction and will dump all the mechanical energy inside the supply.

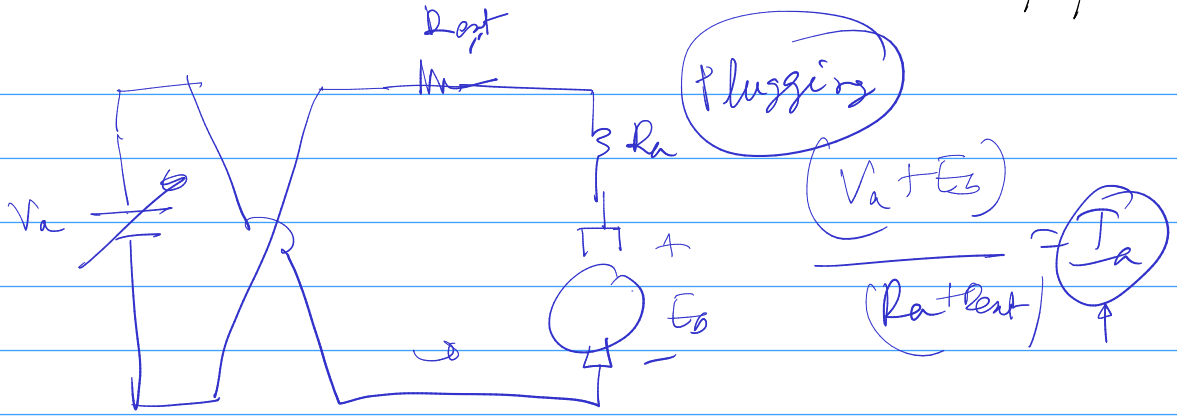
Plugging ::





20th class (19th class Quiz 1)

24/02/2024



D.C. Motor → (1) Separately excited D.C. Motor.

↳ you need 2 separate voltage sources

(2) Shunt motor → (as a generator)

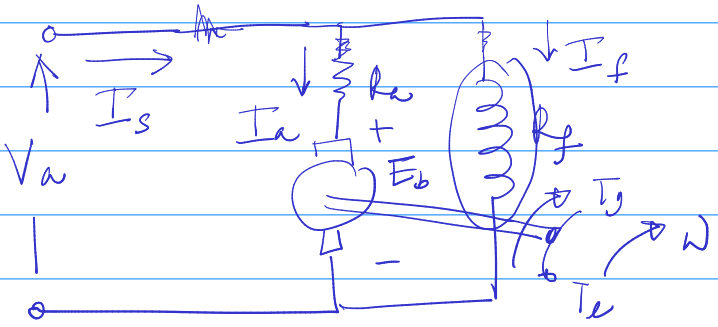
(3) Series motor

Shunt motor

$I_a = 10A$

$I_f = 2A$

$I_f \ll I_a$



$V_a = R_a I_a + K I_f \cdot \omega \rightarrow (1)$

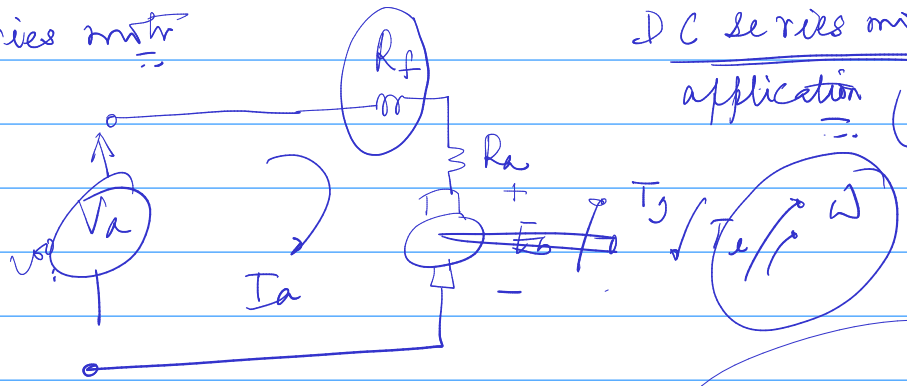
$I_s = I_a + I_f \rightarrow (3)$

$T_g = K I_f \cdot I_a \rightarrow (2)$

$T_g = T_e + J\frac{d\omega}{dt} + B\omega \rightarrow (4)$

Series motor

DC series motor for traction application (starting torque is high)



$T = K\phi I_a$

$V_a = I_a(R_f + R_a) + E_b$

$E_b = K I_a \omega$

$T_g = K I_a^2$

$T_g = T_e + J\frac{d\omega}{dt} + B\omega$

$K I_f \omega + I_a (R_f + R_a) = V_a$

$\Rightarrow \quad \omega = \frac{V_a}{K I_f} - \frac{I_a (R_f + R_a)}{K I_f}$

$T_g = K I_a^2$

$I_a = \sqrt{T_g / K}$

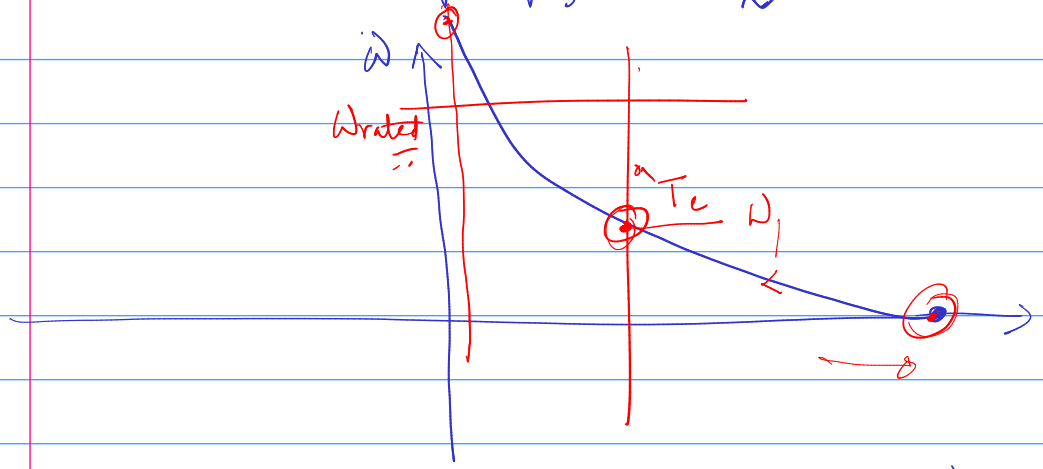
$\omega = \frac{V_a}{\sqrt{K}\sqrt{T_g}} - \frac{(R_f + R_a)}{K}$

$\omega = 0$

$\frac{R_f + R_a}{K} = \frac{V_a}{\sqrt{K}\sqrt{T_g}}$

$\sqrt{T_g} = \frac{\sqrt{K} \cdot V_a}{(R_f + R_a)}$

$T_g = \frac{K V_a^2}{(R_f + R_a)^2}$



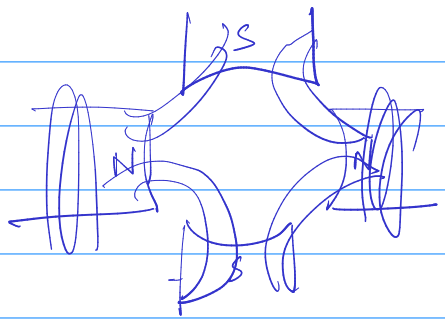
Nameplate: 200V, 4 pole, 1500 rpm, 20 kW, D.C separately excited motor

E_b K

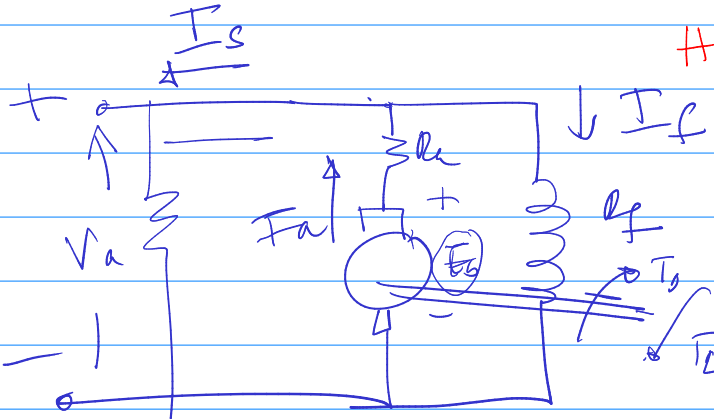
rated speed

rated output power

$T_L \cdot \omega = T_g \cdot \omega = 20\,kW$



D.C. generator → shunt generator



How to start D.C. shunt generator?

$E_b > V_a$

T_e & ω are in the same direction

$T_g \cdot \omega = -ve$

$V_a = E_b - I_a R_a$

$T_g \omega = E_b \cdot I_a$

$E_b = K I_f \cdot \omega$

$I_f = V_a / R_f$

$T_g = K I_f \cdot I_a$

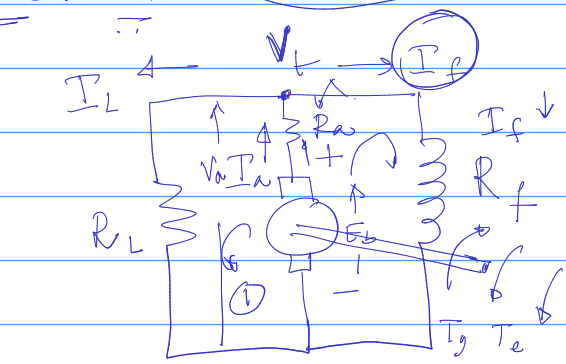
$I_s = I_a - I_f$

21st class

26/02/2026

$0 \leq R_f < \infty$

D.C. Generator



$E_b - I_a R_a = V_a$, $V_a / R_f = I_f$

$V_t = I_f R_f$

I_g, ω are in opposite direction

T_e = external mechanical source.

Wind / Diesel generator / water turbine / steam turbine

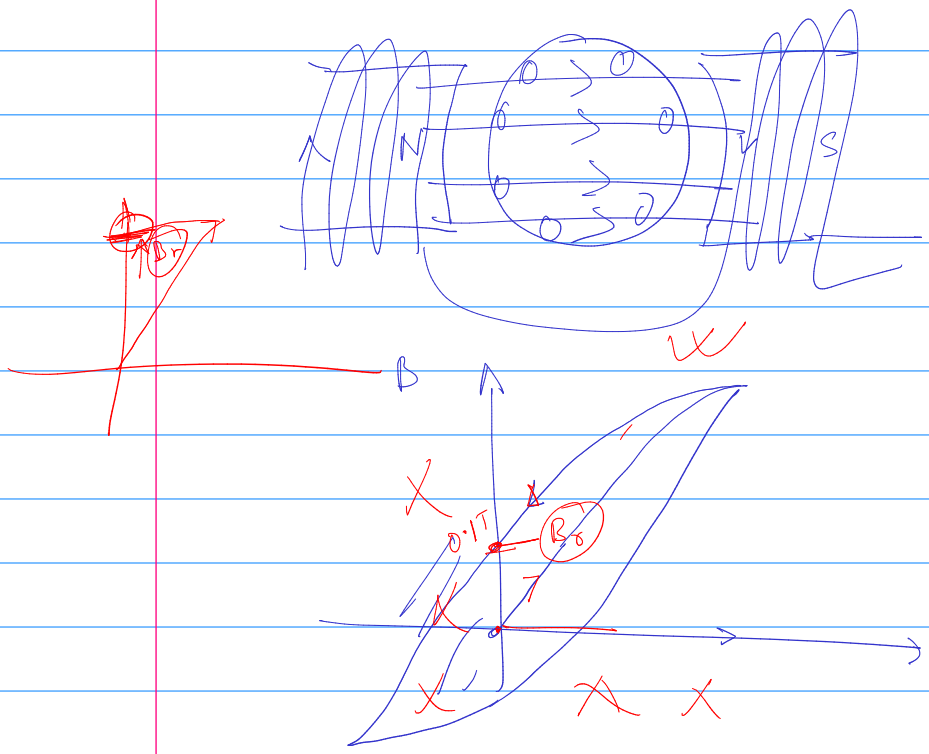
$E_b = I_a R_a + I_a R_L \rightarrow (1)$

$V_a = I_a R_L \rightarrow (2)$; $I_f = (V_a / R_f)$

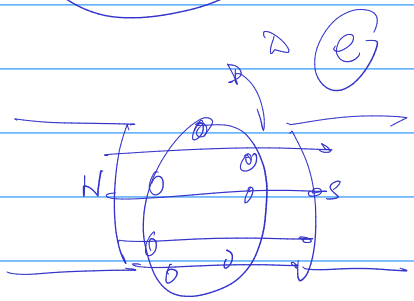
$E_b = K I_f \cdot \omega$. $T_g = K I_f I_a$

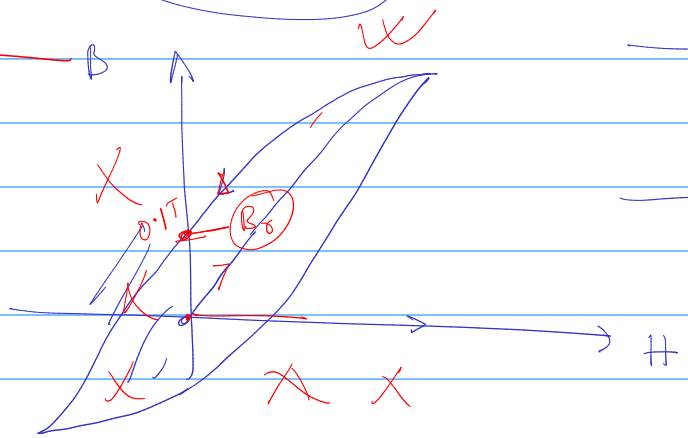
$E_b = k \phi \omega = 0$

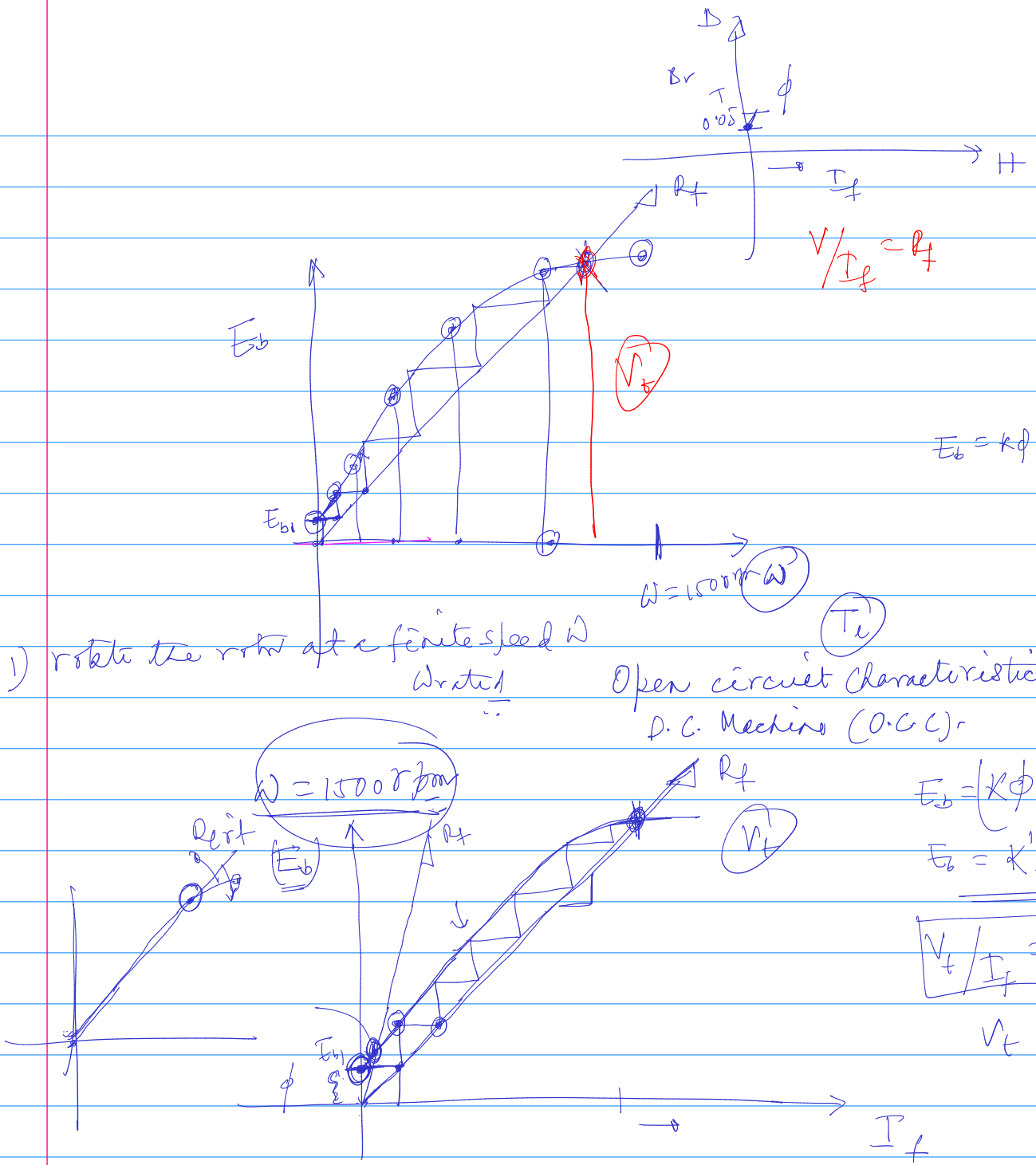
Hence a D.C. generator develops voltage



$I_f = 0$







1) rotate the rotor at a finite speed ω ω_{rated}

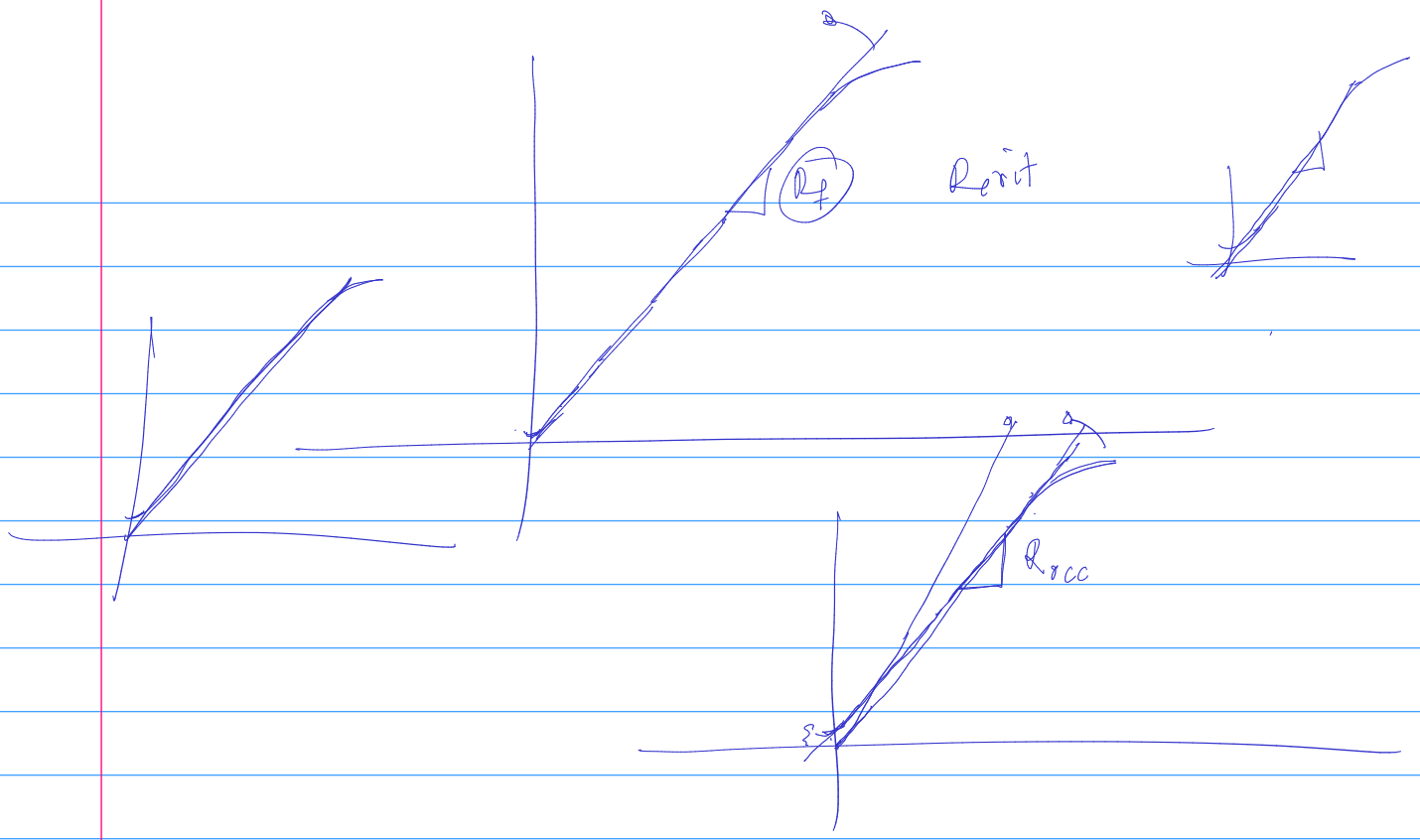
Open circuit characteristics of D.C. Machine (O.C.C).

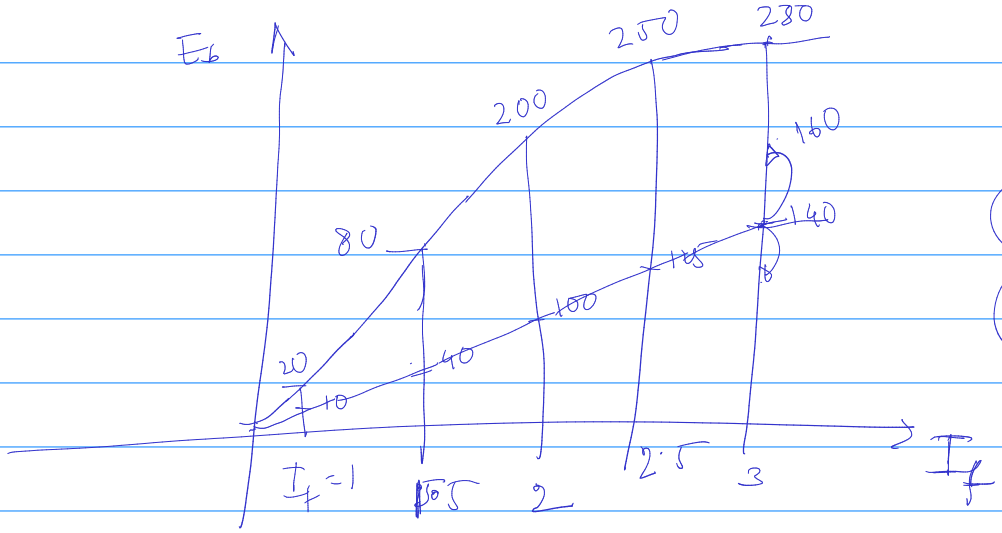
$E_b = (K\phi \cdot \omega_m)$

$E_b = K' I_f \cdot \omega_m$

$\boxed{V_t / I_f = R_f}$

$V_t = R_f I_L$



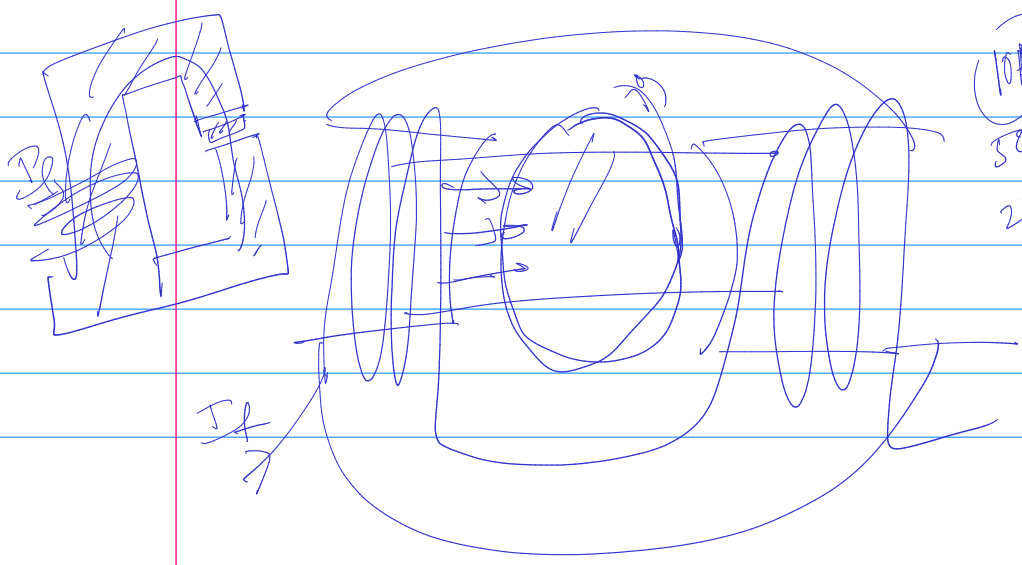


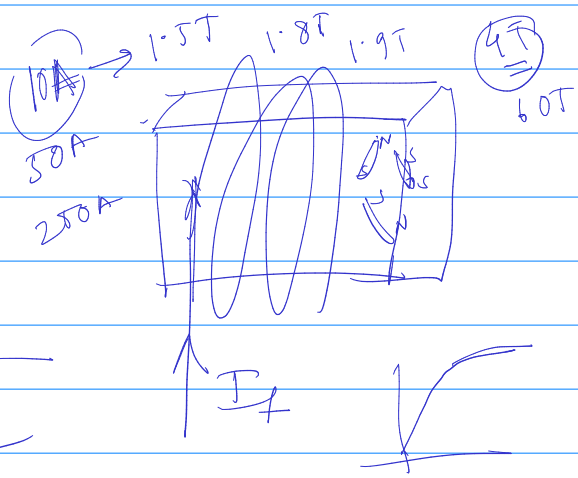
N = 1500 rpm

N = 750 rpm

E_b 1500

E_b 750





22nd class

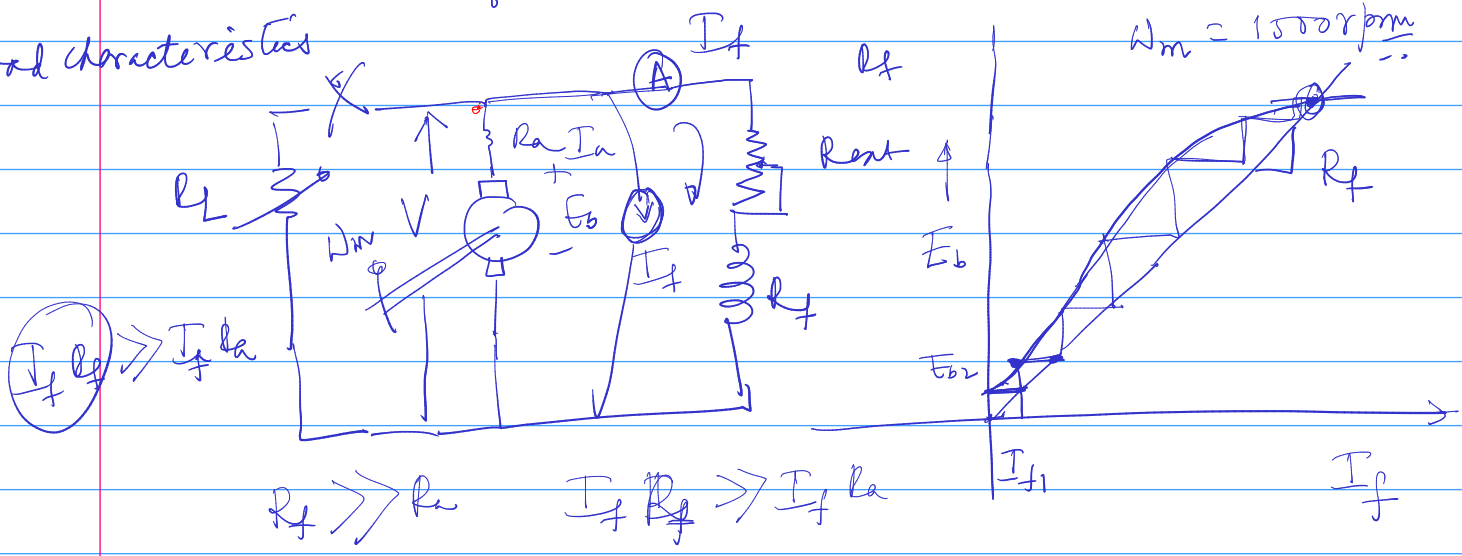
27/02/2026

O.C.C (open circuit characteristics) for D.C shunt generator.

$W_{rated} = 1500$ rpm

$W_m = 1500$ rpm

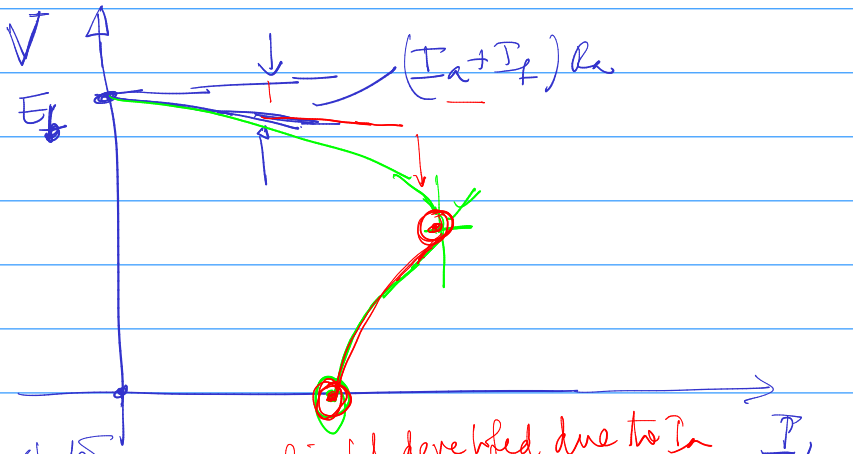
Load characteristics



$V = E_b - (I_a + I_f) R_a$

$I_a \gg I_f$

Load characteristics

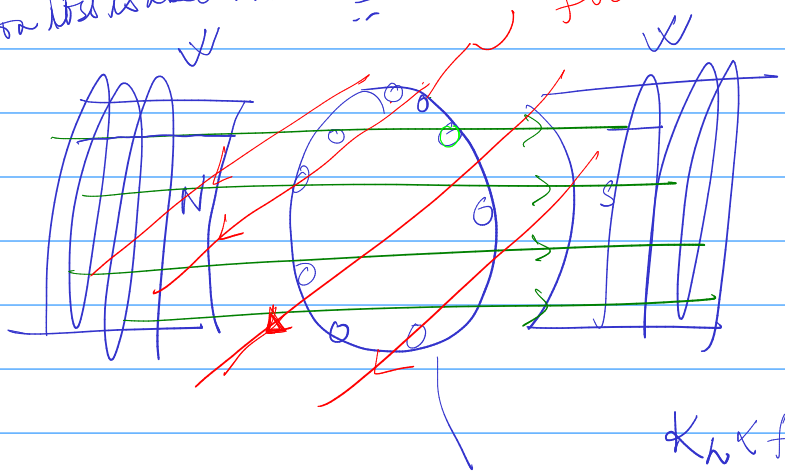


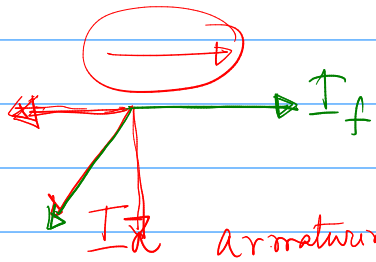
Armature reaction

$E_b = K \phi W_m$

iron loss is absent in stator

field developed due to I_a





armature reaction

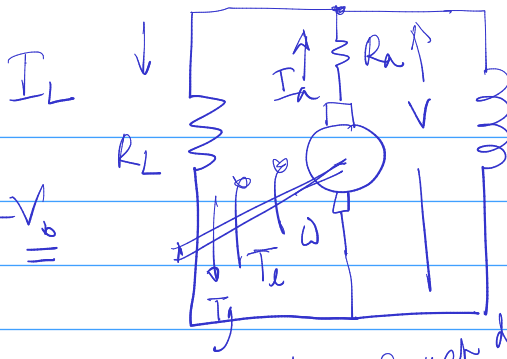
main field is reduced

$K_h \propto f^{1.6}$, $K_e \propto f^2$

rotor (hysteresis + eddy current loss)

Efficiency

$I_a = (I_L + I_f)$



$R_f \gg R_L$

$I_f \ll I_L$

$I_a \simeq I_L$

$E_b = V + I_a R_a + V_b$

$= V_b I_a$

$P_{in} = T_d \omega$

Iron loss, Brush drop

$I_L^2 R_a$

(air gap power) $= E_b I_a$

$V I_L$

$I_f^2 R_f$ fixed loss

Mechanical loss

stray loss = Iron loss + Mechanical loss

= const = fixed loss $= P_f$.

$$\eta\% = \frac{\text{output power}}{\text{Input power}}$$

$$= \frac{V I_L}{(V I_L + I_L^2 R_a + P_f)} \quad \text{Input power}$$

$$= \frac{1}{D = \left(1 + \frac{R_a I_L}{V} + \frac{P_f}{V I_L}\right)}$$

$\eta\%$ will be max D = is min

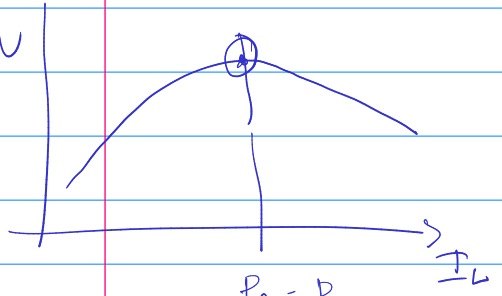
$$\frac{dD}{dI_L} = \frac{R_a}{V} - \frac{P_f}{V I_L^2} = 0$$

$$\frac{P_f}{V I_L^2} = \frac{R_a}{V}$$

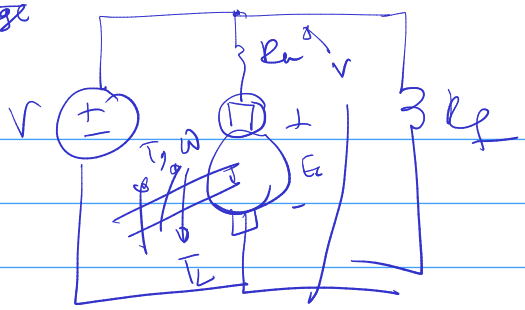
$$\boxed{P_f = R_a I_L^2}$$

(fixed loss = variable loss)

condn for max efficiency



Brush is a lossy component
careful about the voltage
drop!!!!



motor:

$$V = I_a R_a + V_b + E_b$$

$$E_b = V_b + I_a I_a + V$$

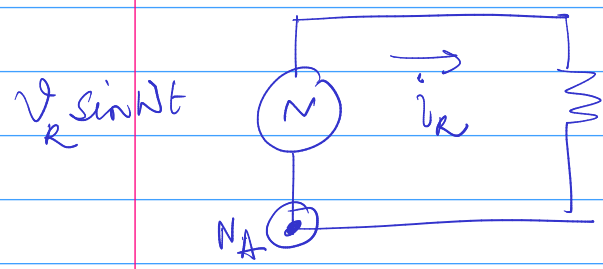
generator:

A.C. Machine

3-ϕ machine = { 1) Synchronous Generator / motor
2) Induction motor / generator

Series A/c circuit / 3-ϕ circuit
↳ Transient + (steady state)

Y / Δ

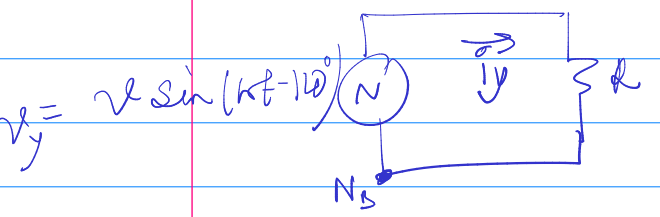


$v_R = V \sin \omega t$

$i = \frac{V \sin \omega t}{R}$

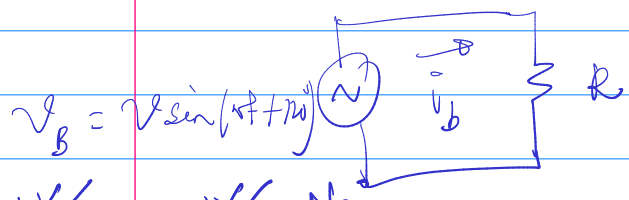
$i = \frac{V}{Z} \sin(\omega t - \phi)$

$i_R = i_m \sin \omega t$



$i_y = \frac{V}{R} \sin(\omega t - 120°)$

$= i_m \sin(\omega t - 120°)$



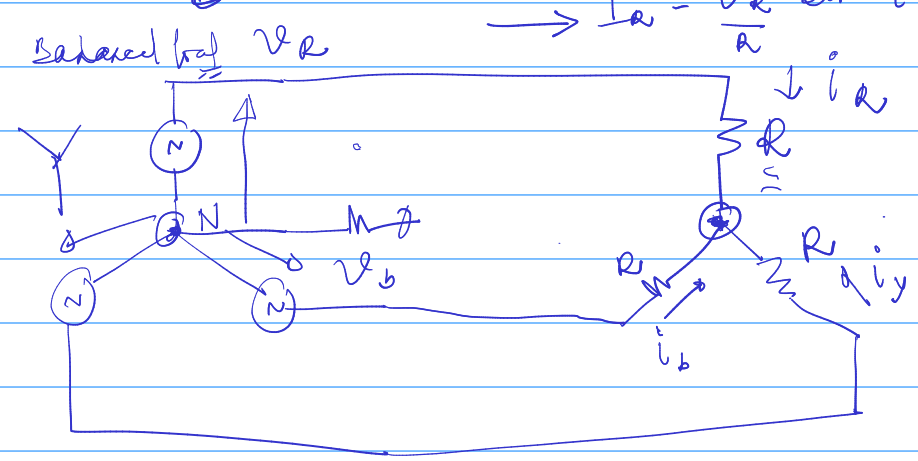
$i_b = \frac{V}{R} \sin(\omega t + 120°)$

Balanced supply

Y

Balanced load V_R

$\rightarrow I_R = \frac{V_R}{R} \sin \omega t = i_R$

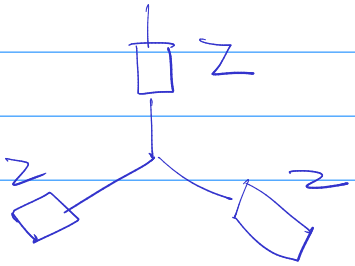


v, ω, and phase = 120°

$i_N = \frac{V}{R} \sin \omega t + \frac{V}{R} \sin(\omega t - 120°) + \frac{V}{R} \sin(\omega t + 120°)$

$= 0$

$Z = R + j\omega L$



24th class (23rd DC Machine Tutorial)

03/03/2026

3ϕ AC Circuit

Balanced 3-ϕ supply

$V_R = V_{mR} \sin \omega t$

$V_Y = V_{mY} \sin(\omega t - 120^\circ)$

$V_B = V_{mB} \sin(\omega t + 120^\circ)$

1) $V_{mR} = V_{mY} = V_{mB}$

2) ω is also fixed.

3) Phase difference amongst all the phases are maintained at 120°.

Balanced load $\quad Z_R = Z_Y = Z_B$

$(R_L + j\omega L_L) = R_Y + j\omega L_Y = R_B + j\omega L_B$

$R_R = R_Y = R_B \qquad L_R = L_Y = L_B$

Balanced supply + Balanced load

$i_R = i_m \sin \omega t$

$i_Y = i_m \sin(\omega t - 120^\circ)$

$i_B = i_m \sin(\omega t + 120^\circ)$

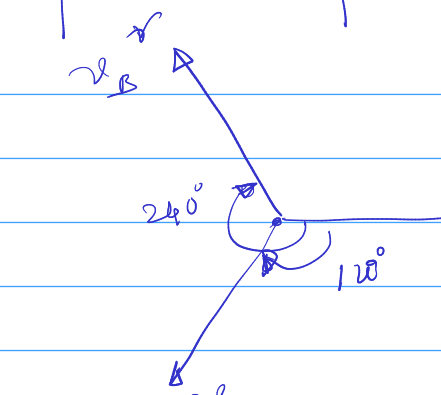
$i_R + i_Y + i_B = 0$

Y — Y

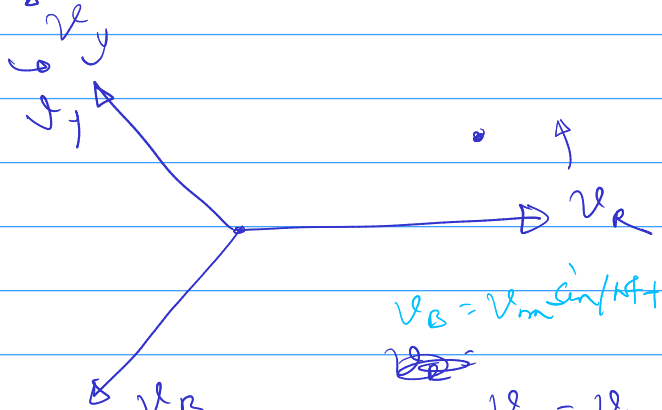
$v_R = V_m \sin \omega t$

$v_y = V_m \sin(\omega t + 120^\circ)$

$v_B = V_m \sin(\omega t + 120^\circ)$



RYB phase sq



$v_R = V_m \sin \omega t$

$v_B = V_m \sin(\omega t - 120^\circ)$

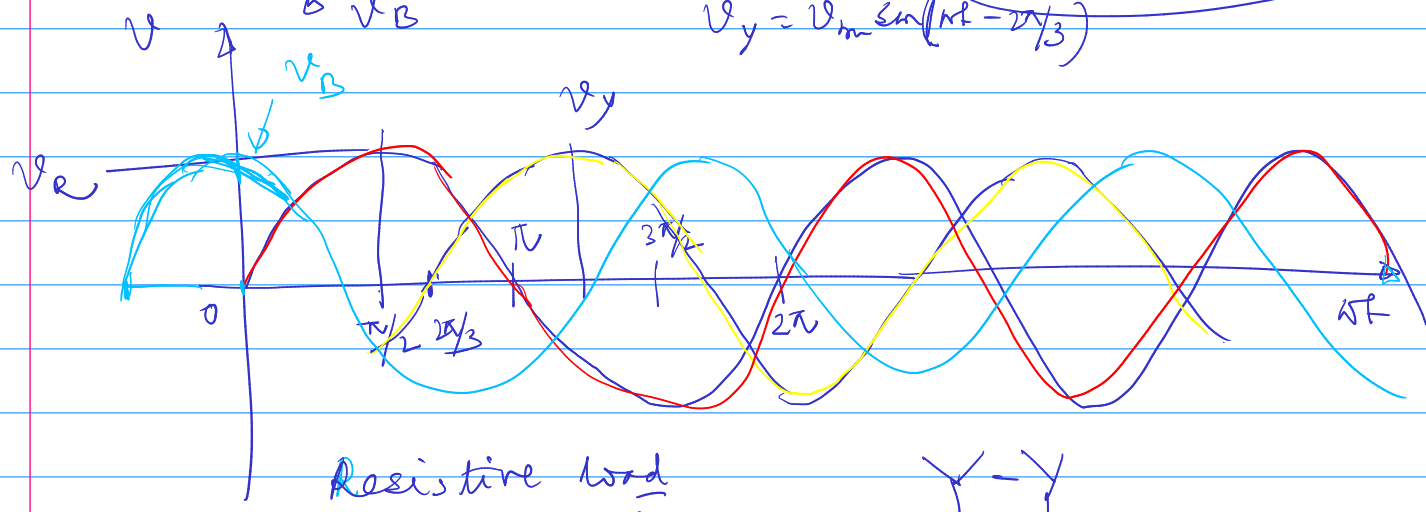
$v_y = V_m \sin(\omega t + 120^\circ)$

$v_B = V_m \sin(\omega t + 2\pi/3)$

RBY phase sq

$v_y = V_m \sin(\omega t - 2\pi/3)$

RYB



Resistive load

Y — Y

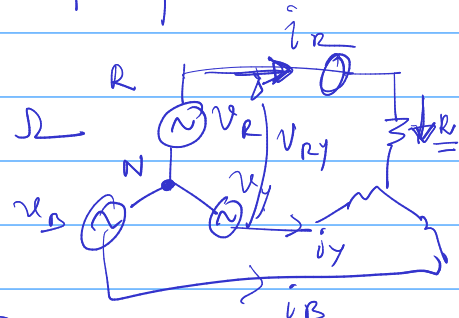
$\omega = 2\pi \times 50$

$v_R = 100 \sin \omega t$

$v_y = 100 \sin(\omega t - 2\pi/3)$

$v_B = 100 \sin(\omega t + 2\pi/3)$

$R = 10\ \Omega$

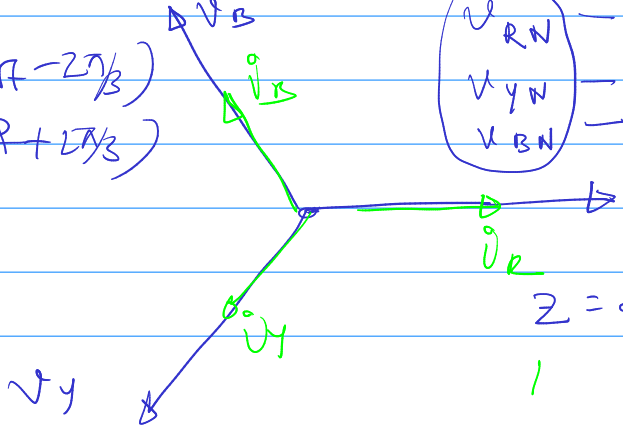


$i_R = \frac{100}{10} \sin \omega t = 10 \sin \omega t$

$i_y = 10 \sin(\omega t - 2\pi/3)$

$i_B = 10 \sin(\omega t + 2\pi/3)$

v_{RN}, v_{YN}, v_{BN} — phase voltage



$Z = R \angle \pi (10 + j10)$

$|i_R| = |i_Y| = |i_B| =$

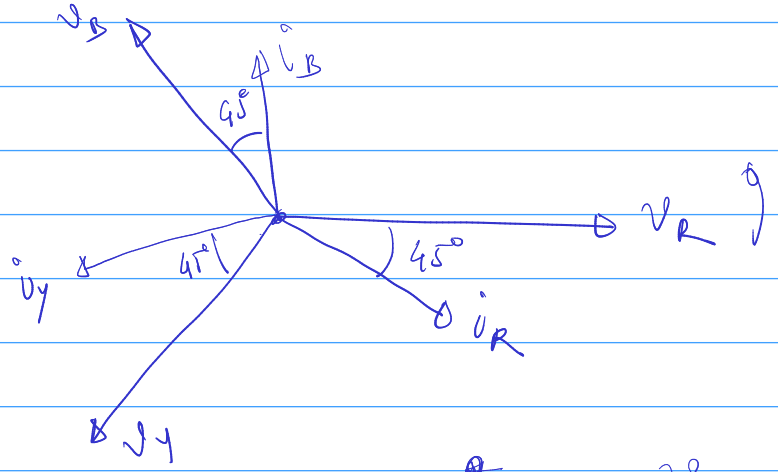
$|Z| = \sqrt{10^2+10^2}$

$= 10\sqrt{2}\angle 45^\circ$

$i_R = \frac{10}{\sqrt{2}} \sin(\omega t - 45^\circ)$

$i_Y = 10/\sqrt{2} \sin(\omega t - 45^\circ - 120^\circ) = 10/\sqrt{2} \sin(\omega t - 165^\circ)$

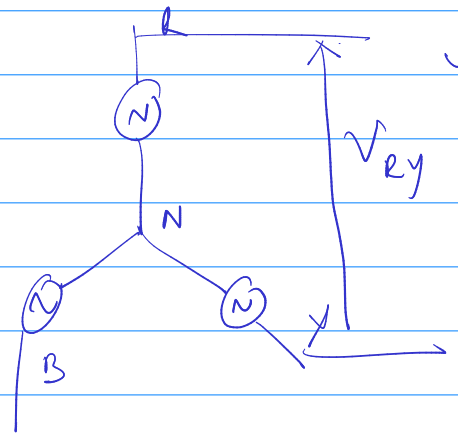
$i_B = 10/\sqrt{2} \sin(\omega t + 165^\circ)$



V_{RY}

V_{YB}

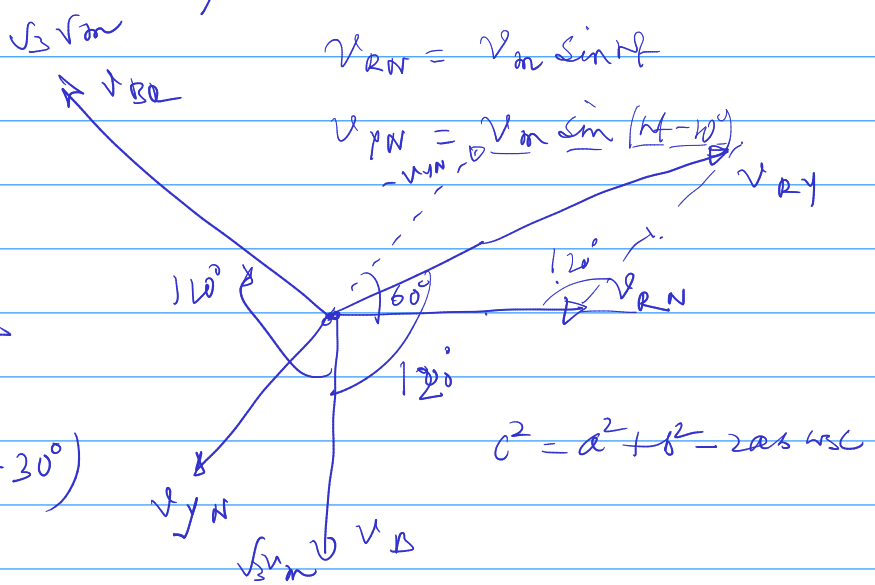
V_{BR}



$V_{RY} = V_{RN} - V_{YN}$

$V_{RN} = V_m \sin\omega t$

$V_{YN} = V_m \sin(\omega t - 120^\circ)$



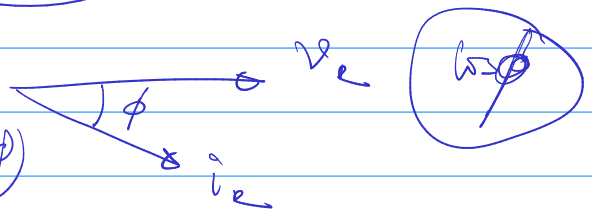
$V_{RY} = \sqrt{3} V_m \sin(\omega t + 30^\circ)$

$c^2 = a^2 + b^2 - 2ab\cos C$

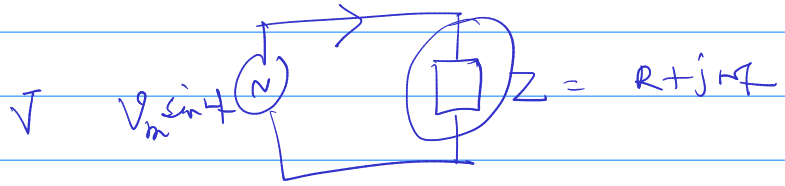
$Y - Y = i_L = i_{ph}$

power factor ϕ =

$I\ i_m(\sin\omega t - \phi)$



$\omega - \phi$



$P = Re(V I^*)$

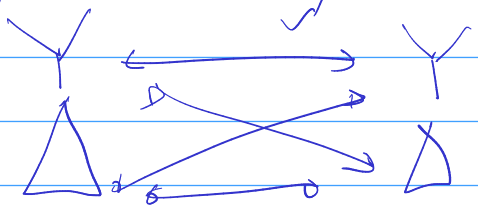
$P = Re[V \cdot I\angle -\phi]^*$

$= Re[V \cdot I\angle \phi]$

3-ϕ ckt power consumed

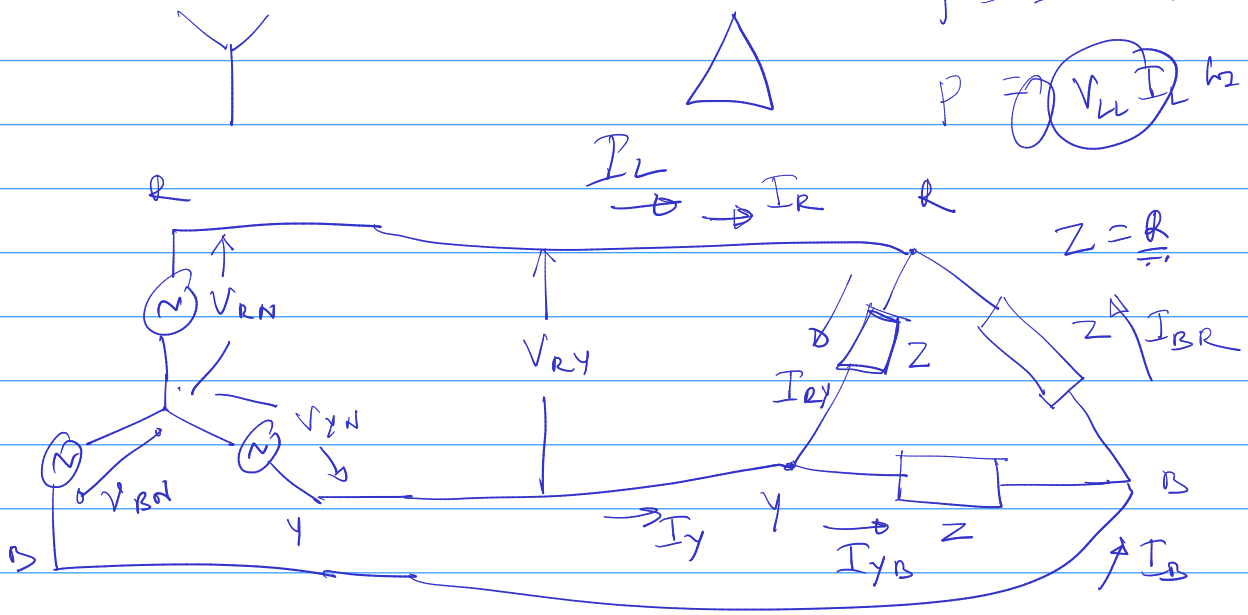
$P = V I \cos\phi$

$P_{3\phi} = 3 V_{ph} I_{ph} \cos\phi = 3 \cdot \frac{V_{LL}}{\sqrt{3}} \cdot I_L \cos\phi = \sqrt{3} V_L I_L \cos\phi$



$P = 3 V_{ph} I_{ph}$

$P = V_{LL} I_L \cos\phi$



$I_{RY} = V_{RY}/Z \quad I_{YB} = V_{YB}/Z \quad , \quad I_{BR} = V_{BR}/Z$

$V_{RR} = V_{LLL}$

$I_R = (\sqrt{3} I_{RY}) \angle -30$

$I_Y = \sqrt{3} I_{YB} \angle -30^\circ$

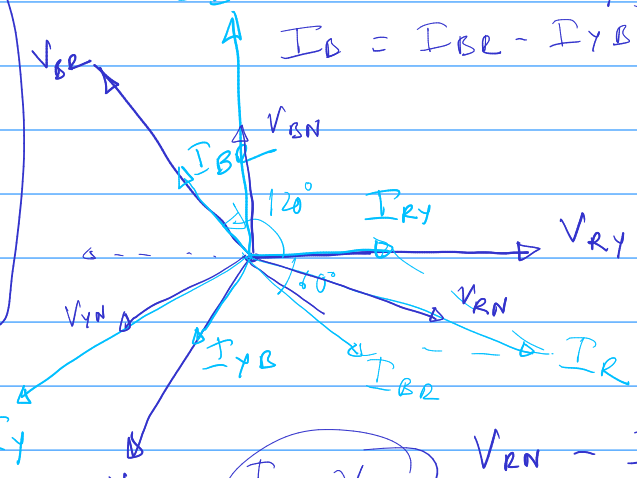
$I_B = \sqrt{3} I_{BR} \angle -30^\circ$

$I_R = I_{RY} - I_{BR}$

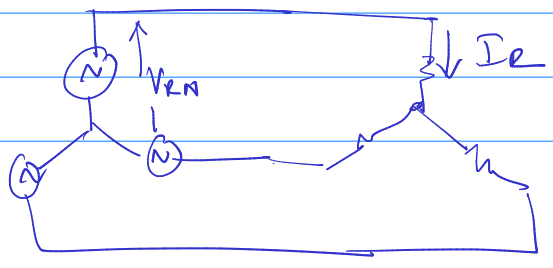
$I_Y = I_{YB} - I_{RY}$

$I_B = I_{BR} - I_{YB}$

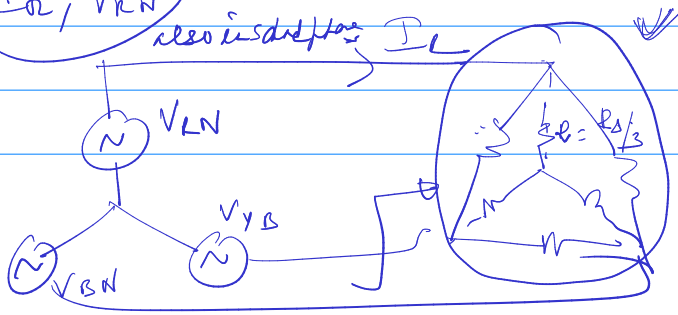
Resistive ckt



V_{RN} – I_R in same phase



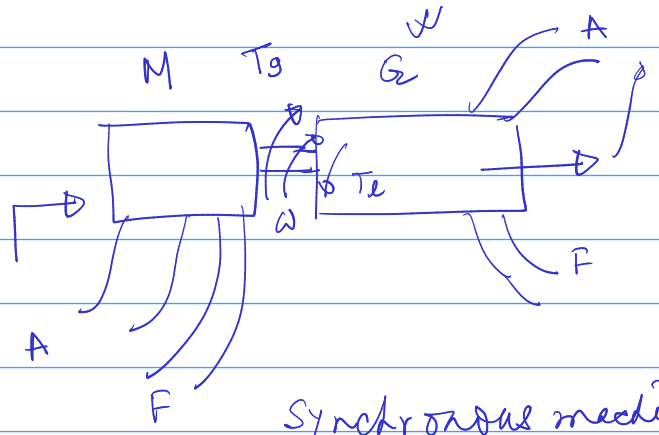
I_R, V_{RN} also in diff phase



25th class

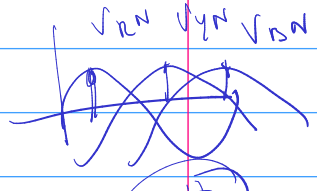
3-ϕ circuit

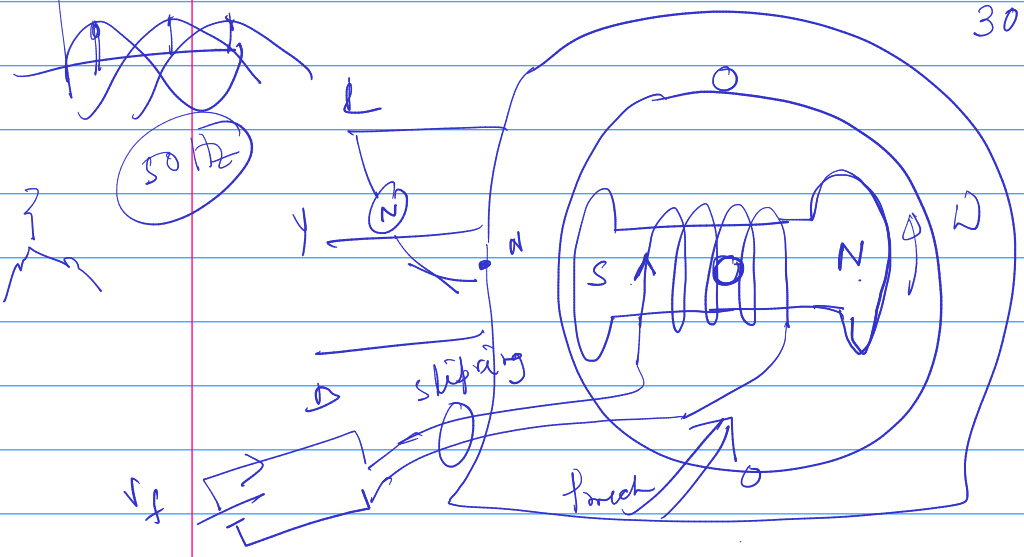
05/03/2026

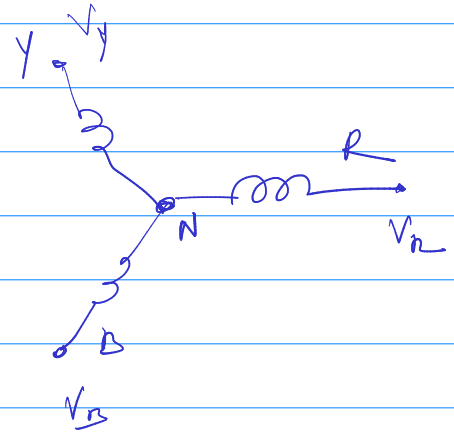


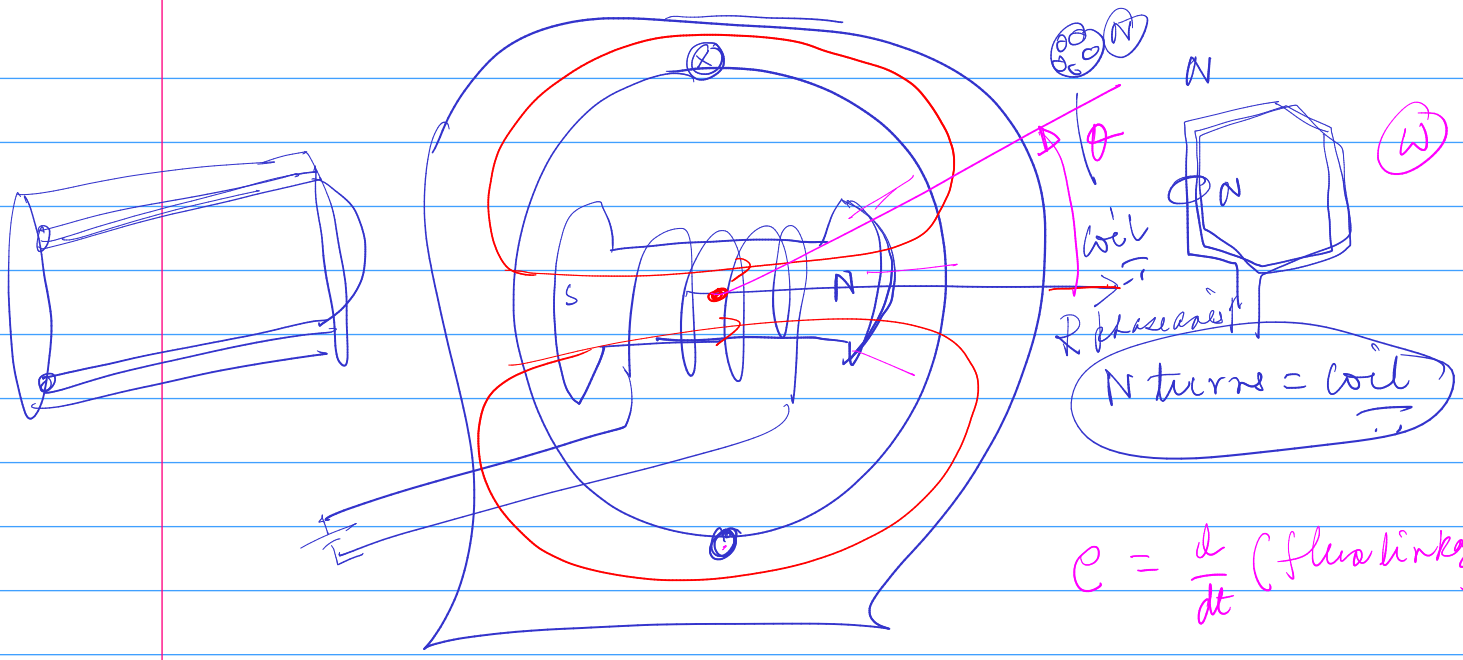
Synchronous machine (Generator)

3000 rpm

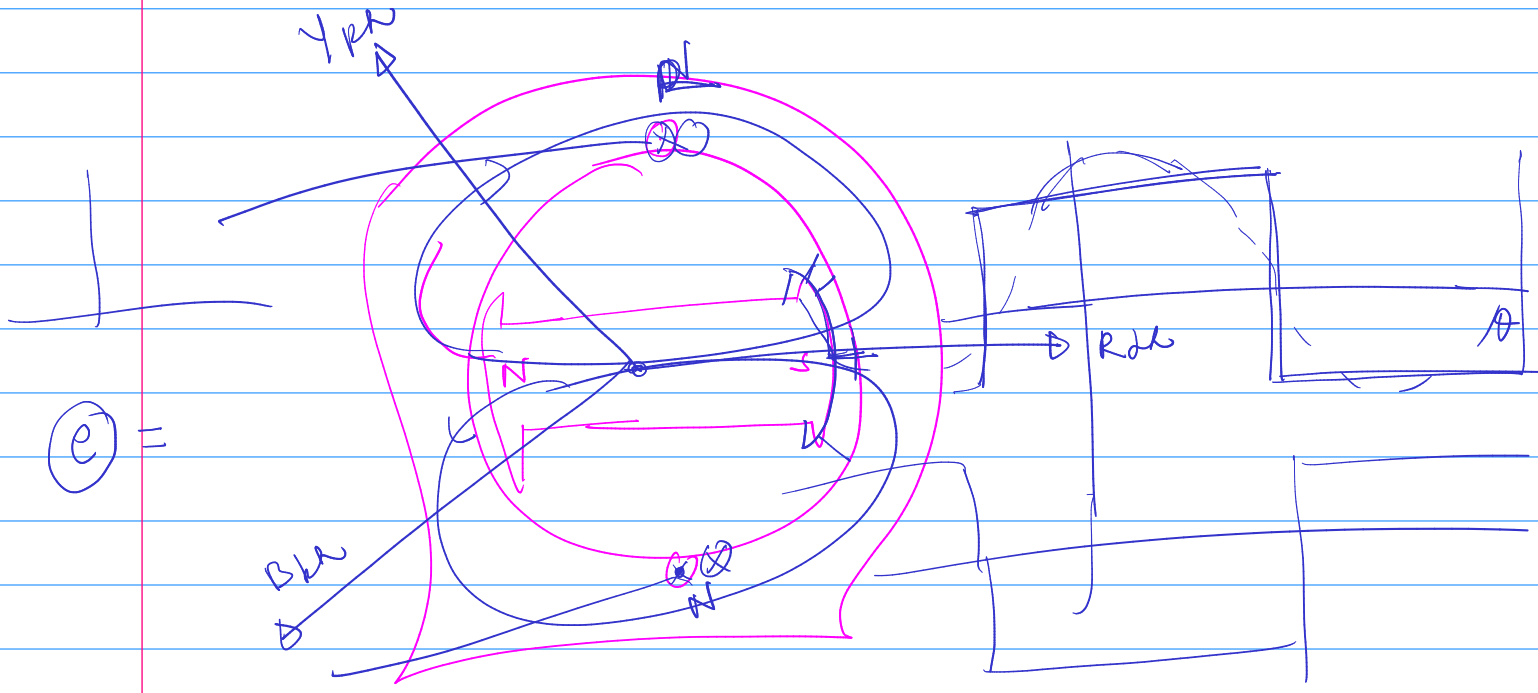




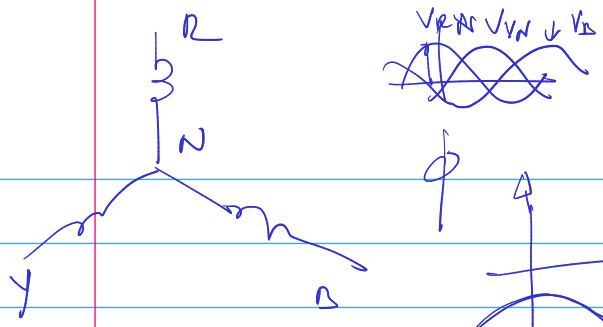


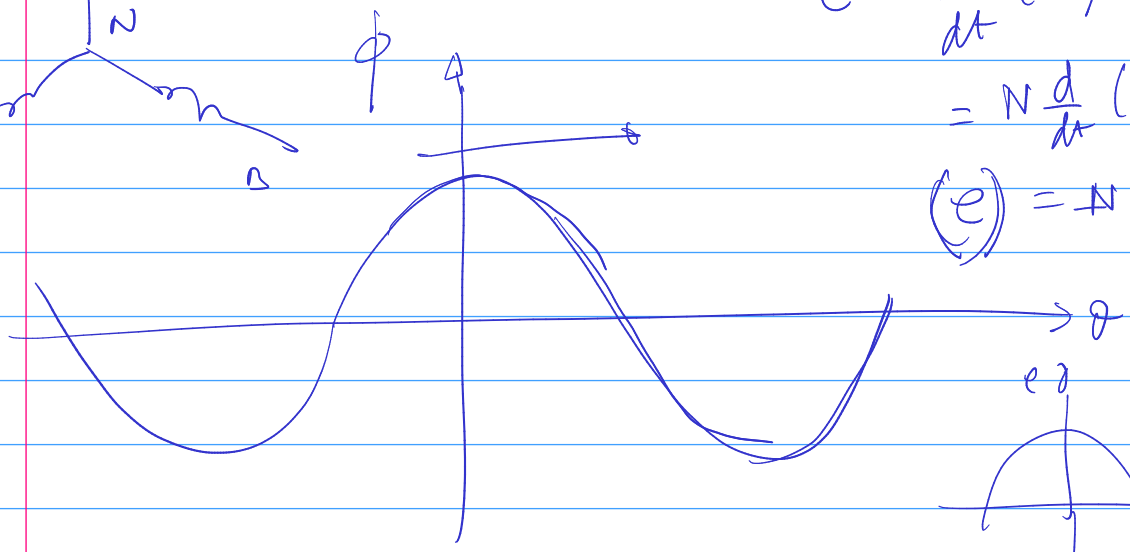


$e = \frac{d}{dt}$ (flux linkage)



e =

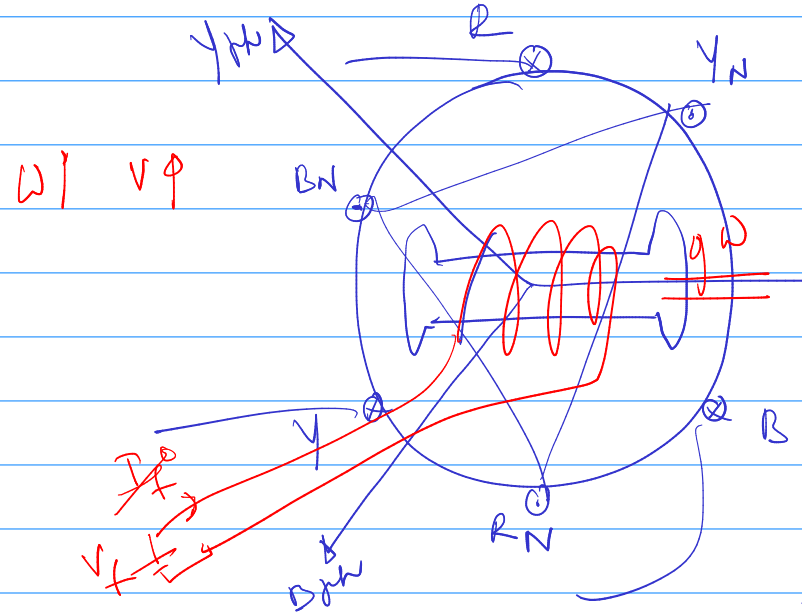


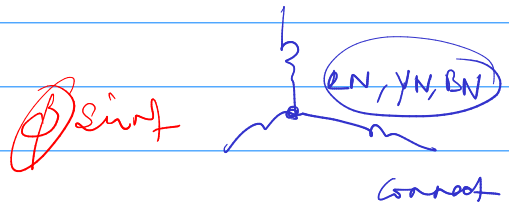


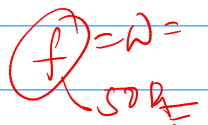
$$e = \frac{d}{dt}(N\phi)$$

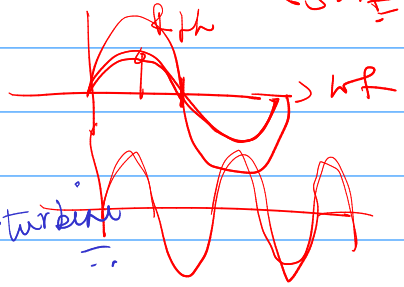
$$= N\frac{d}{dt}(\phi_m \cos\omega t)$$

$$e = N\phi_m \omega \sin\omega t$$



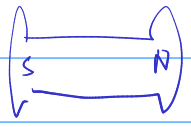






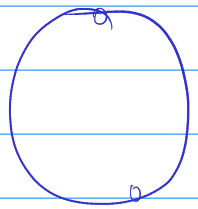
Synchronous m/c → salient pole

→ non salient pole

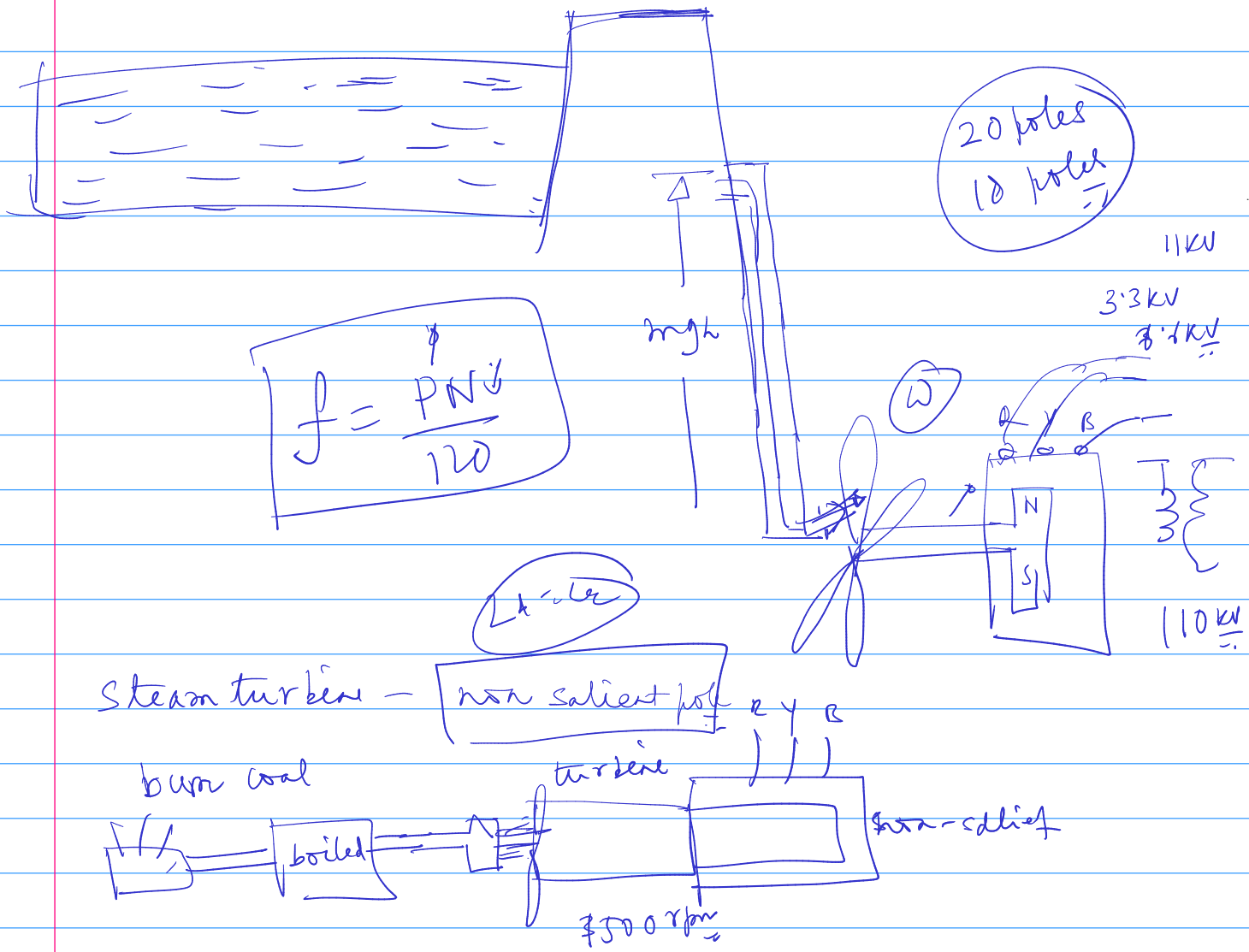


↓

Steam turbine



Hydro turbine

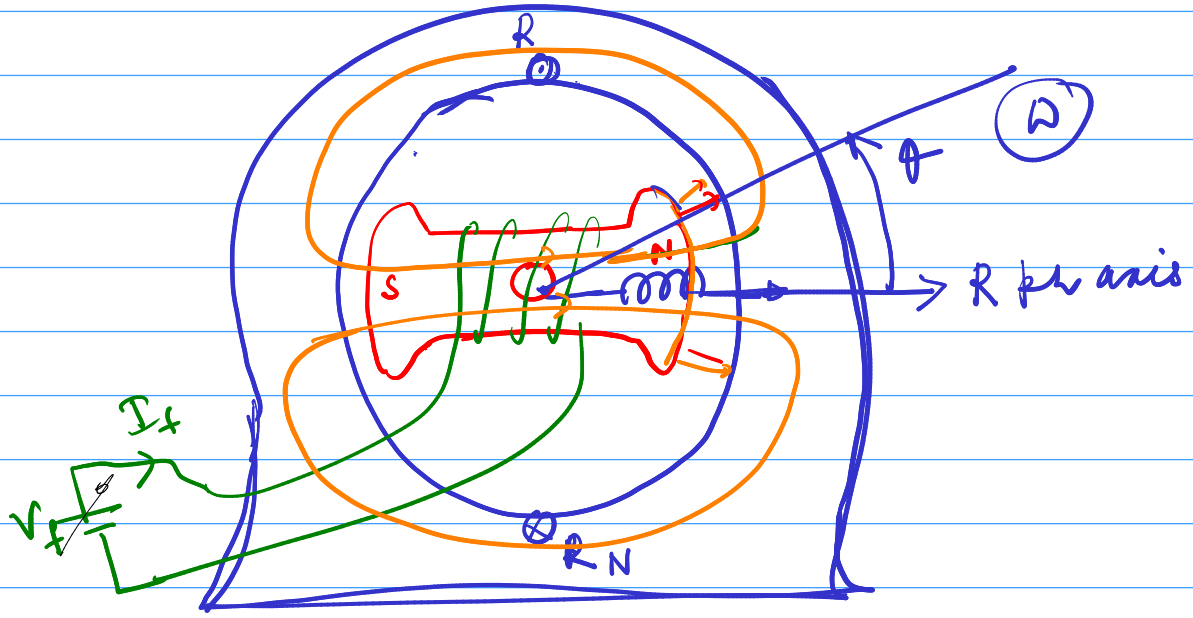


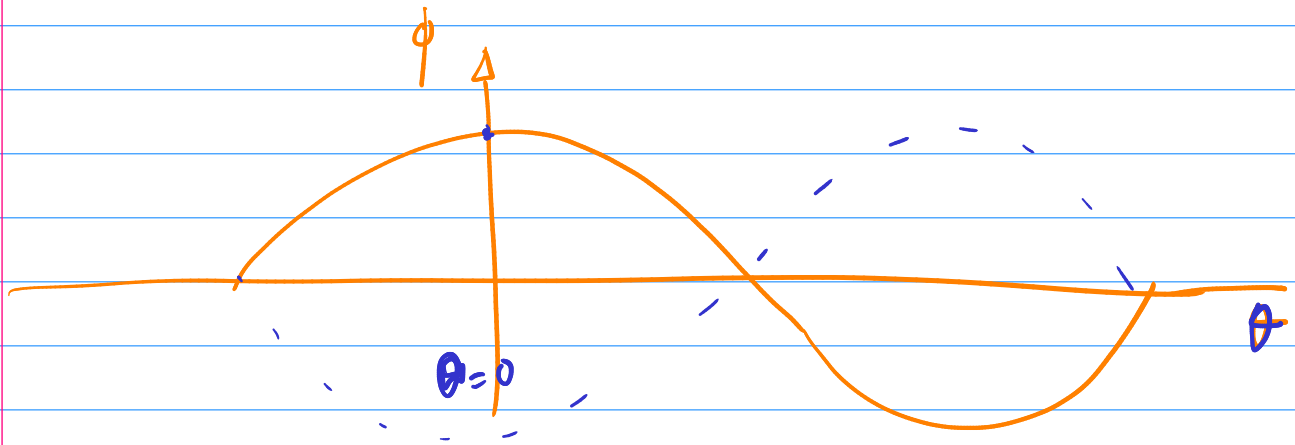
Steam turbine – non salient pole

26th class

Synchronous machine

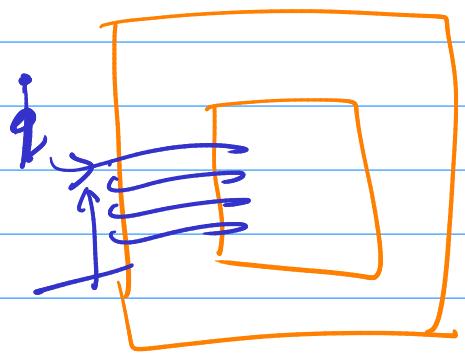
06/03/2026





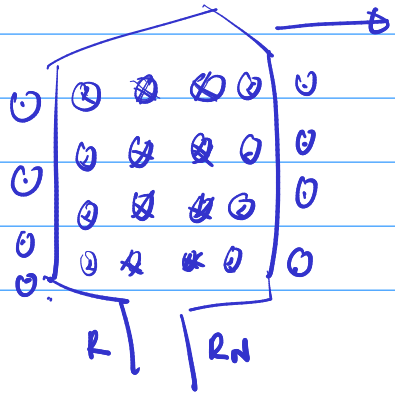
$$e = -\frac{d}{dt}(N\phi) \qquad \phi = \phi_m \cos\omega t$$

$$e = \frac{d}{dt}(N\phi) = +N\phi_m \omega \sin\omega t$$

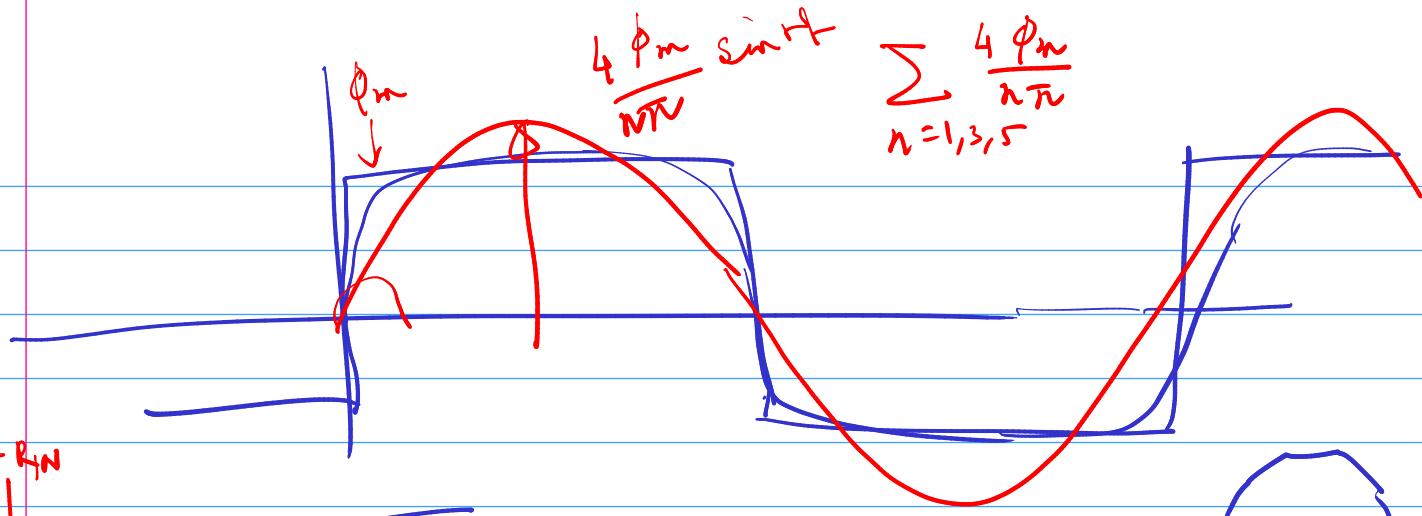


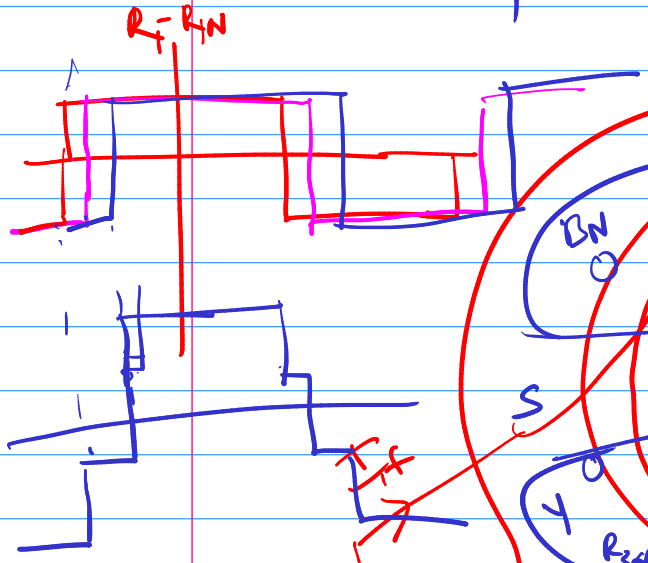
$$\phi = \phi_m \cos\omega t \qquad i_m = i_m \sin\omega t$$

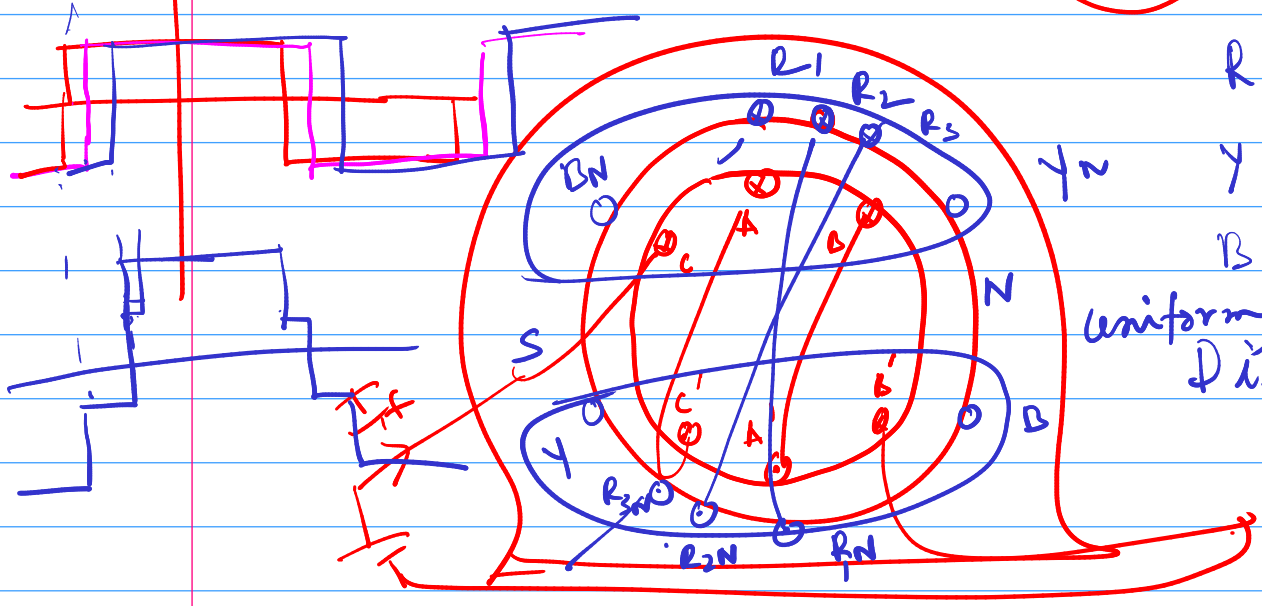




Rotational emf



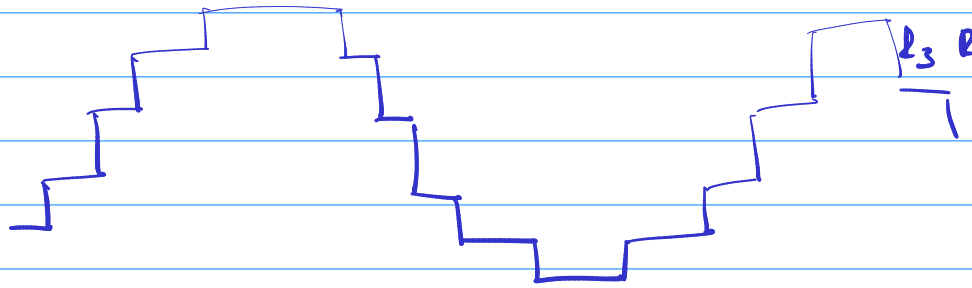




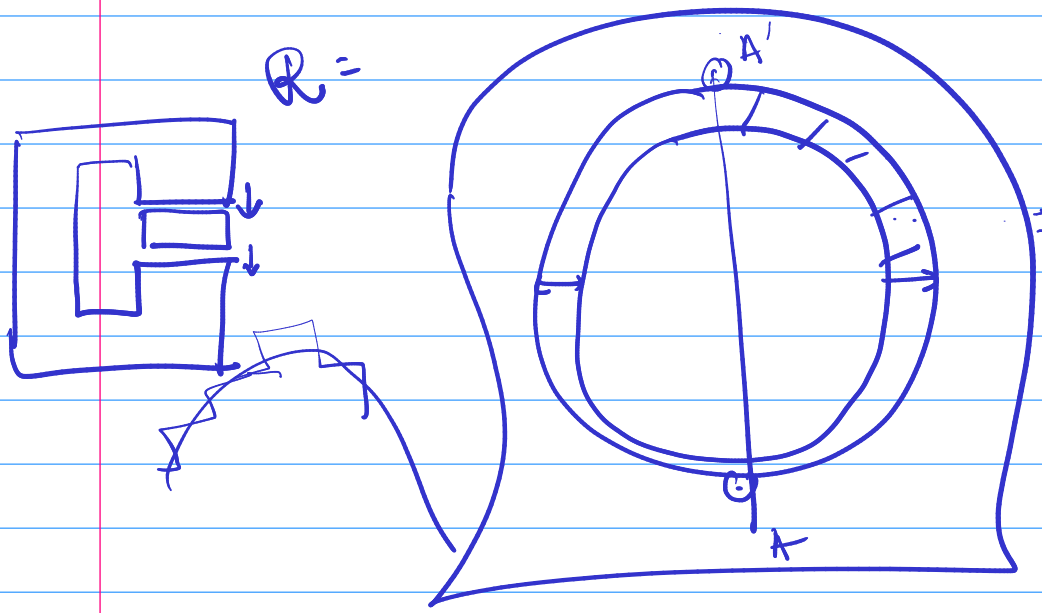
$R R_N$ —
$Y Y_N$
$B B_N$

uniformly Distributed winding

$R_1 R_{1N} = N$
$R_2 R_{2N} = N$
$R_3 R_{3N} = N$



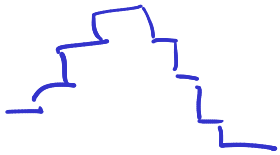
$R =$



mmf

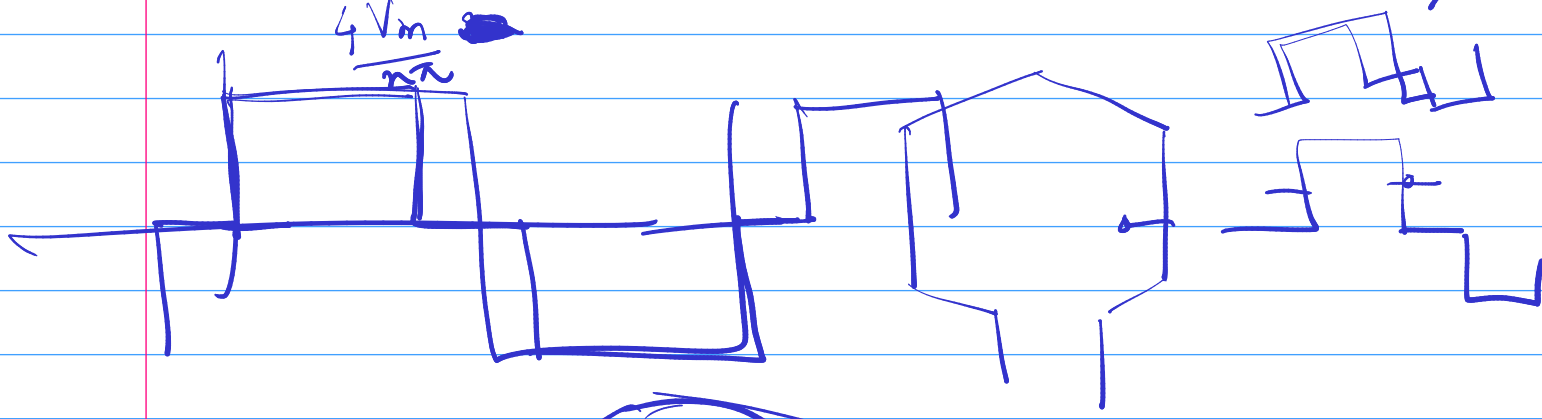


$v = v_m$



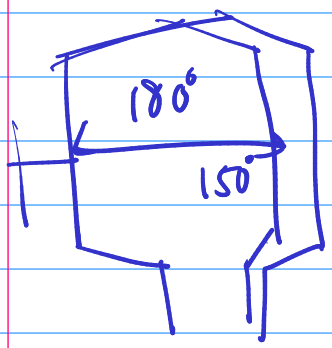
→ sine wave

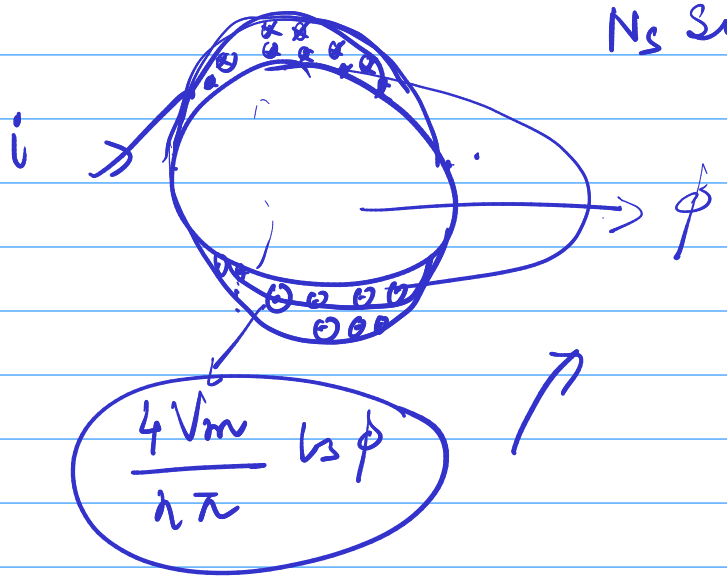
ideal coil windings → sinusoidally distributed in the stator periphery,

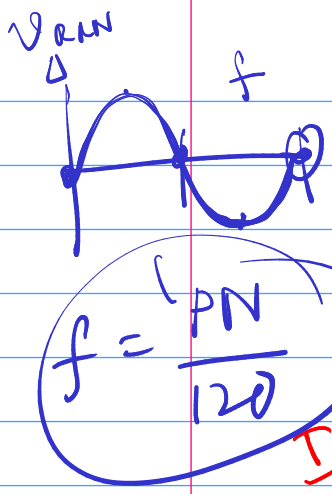




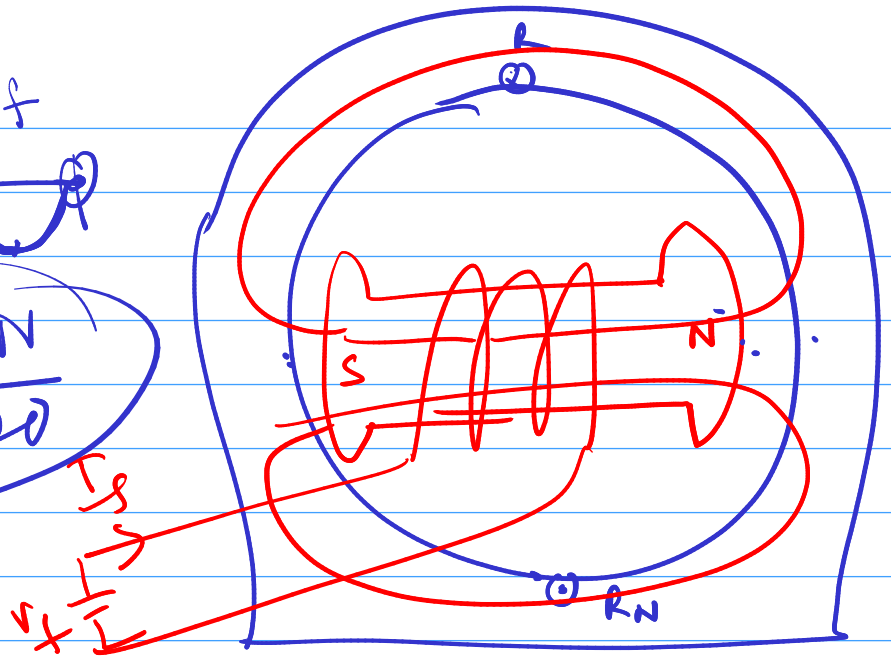
$N_s \sin\phi$







$$f = \frac{PN}{120}$$



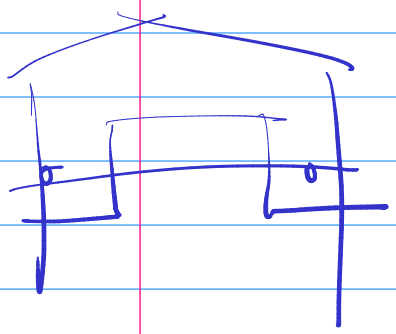
N RPM

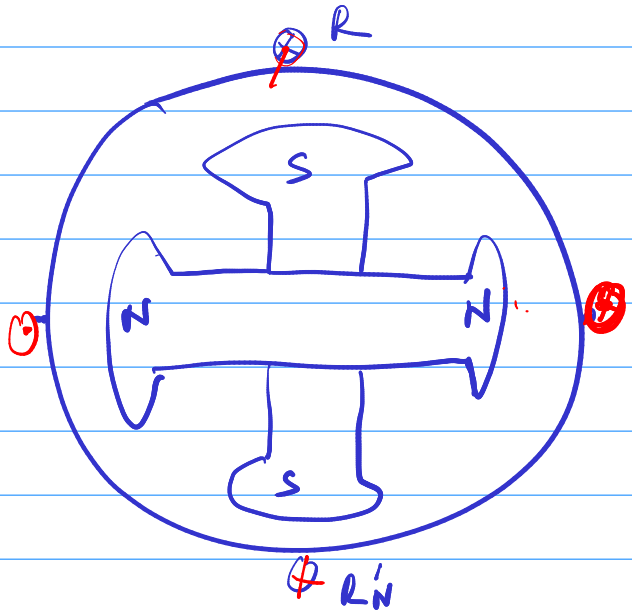
$= 3000$ RPM ∴

n (RPS)

$= \frac{3000}{60}$

$f = \frac{P}{2} \frac{N}{60} = 50$ Hz ∴



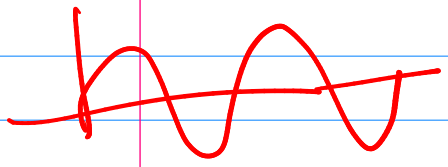


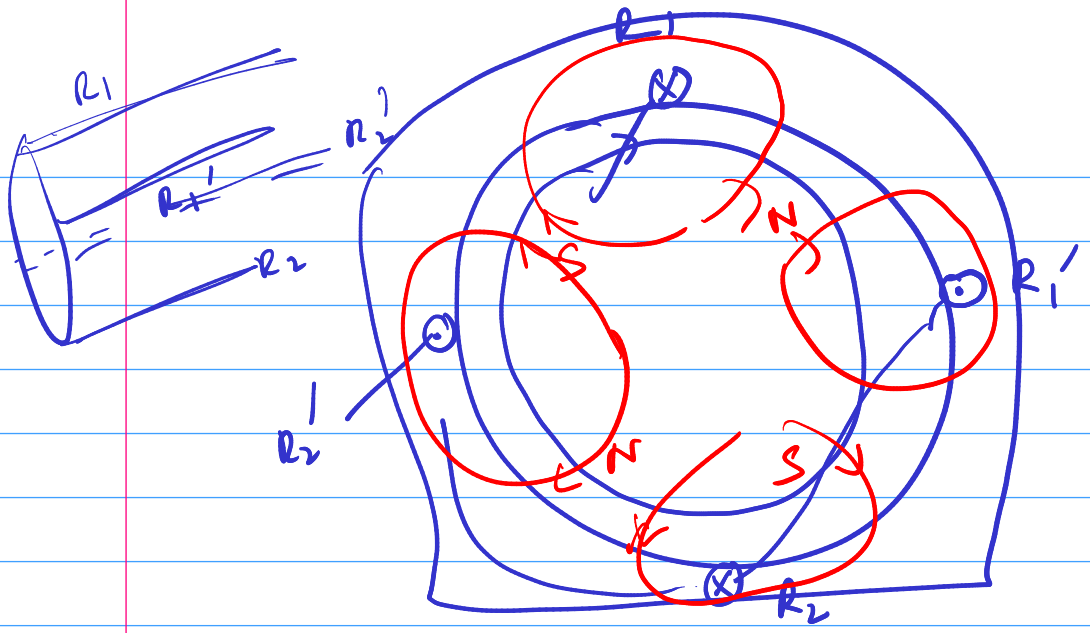
3000 r/m ∴

$f = \frac{P}{2} \cdot \frac{N}{60}$

$= \frac{4}{2} \cdot \frac{3000}{60}$ 50

$= 100$ r/m ∴





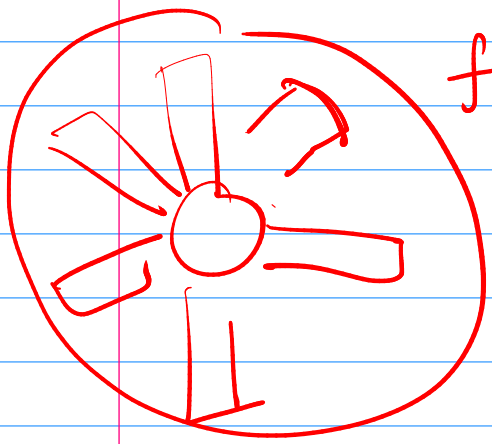
P = 8 kole

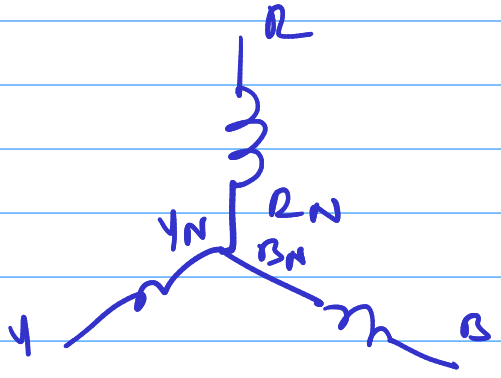
f = 50 Hz

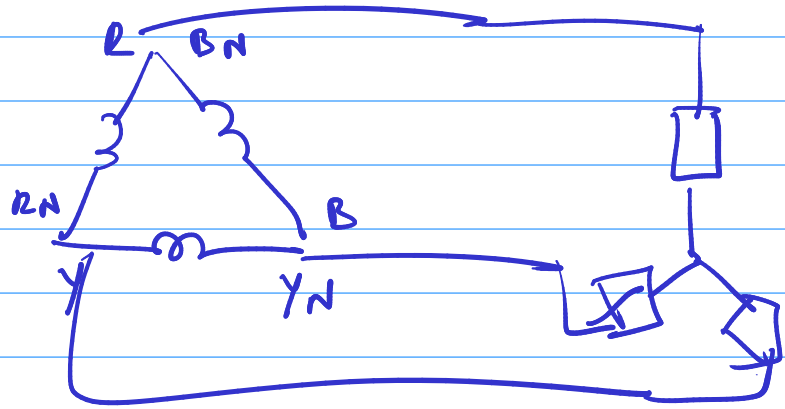
N' =

$f = \frac{PN}{120}$

N = 750 rpm





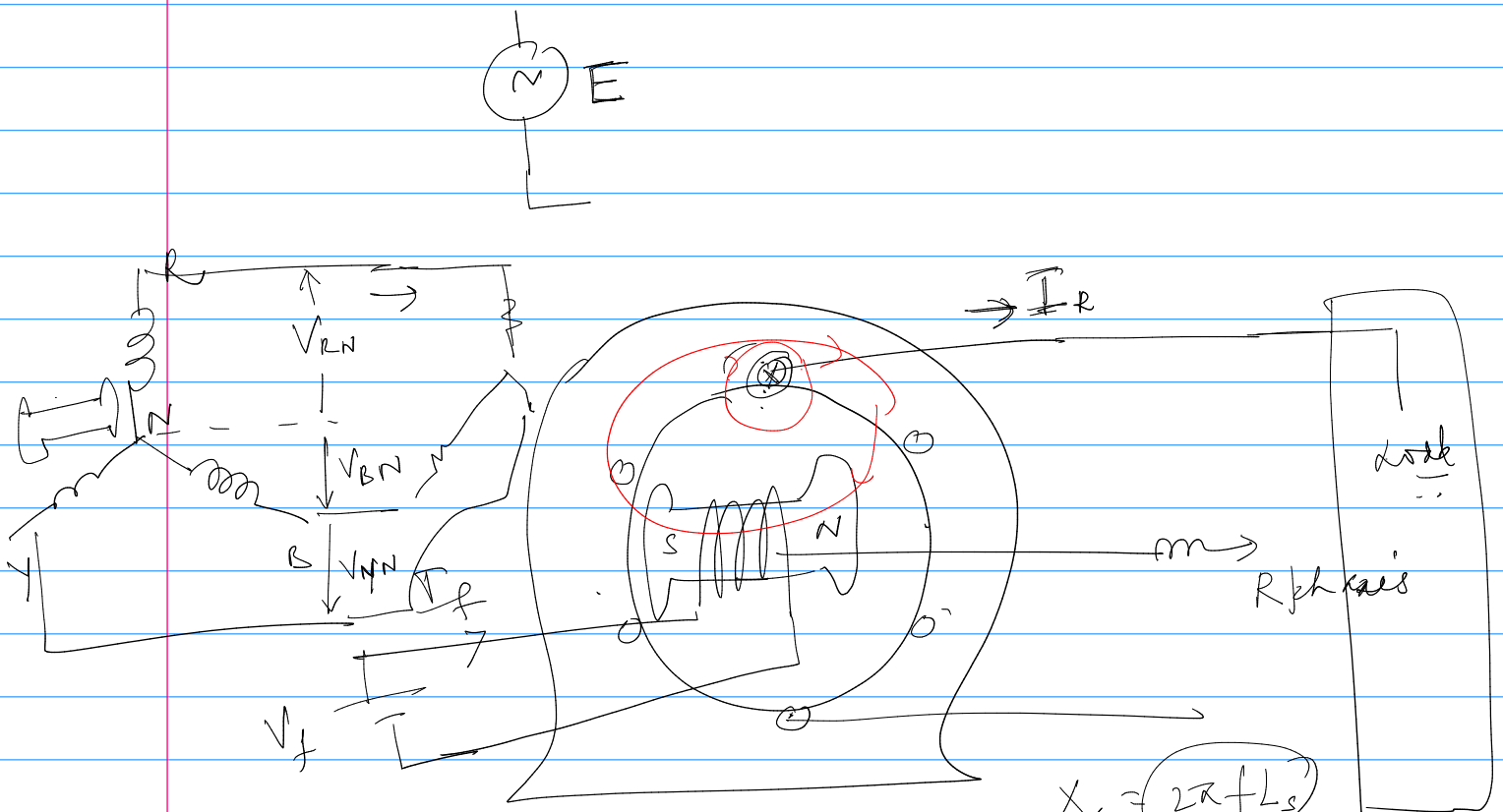


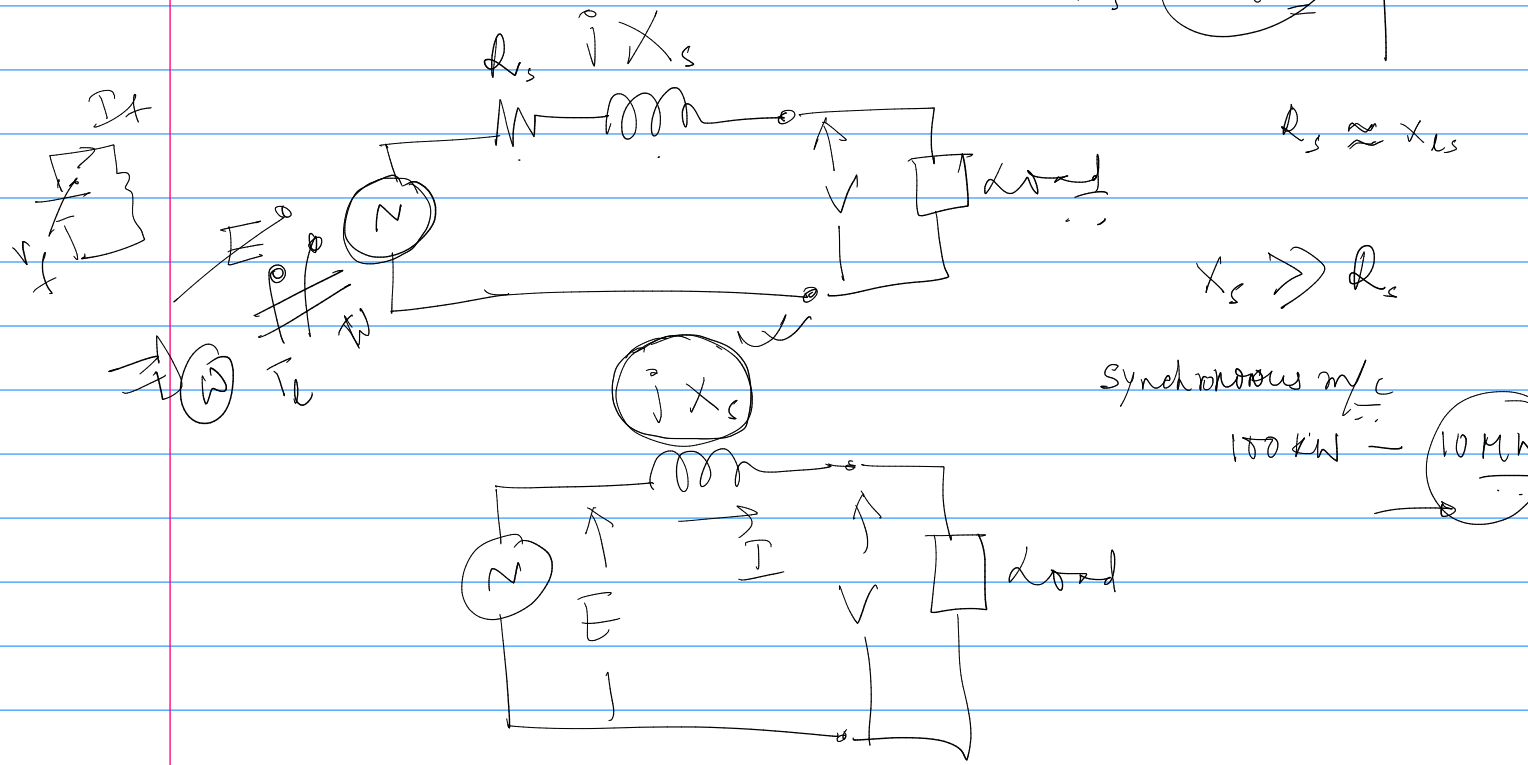
27th class

09 Mar 2026

All the windings will generate voltage of similar magnitude.

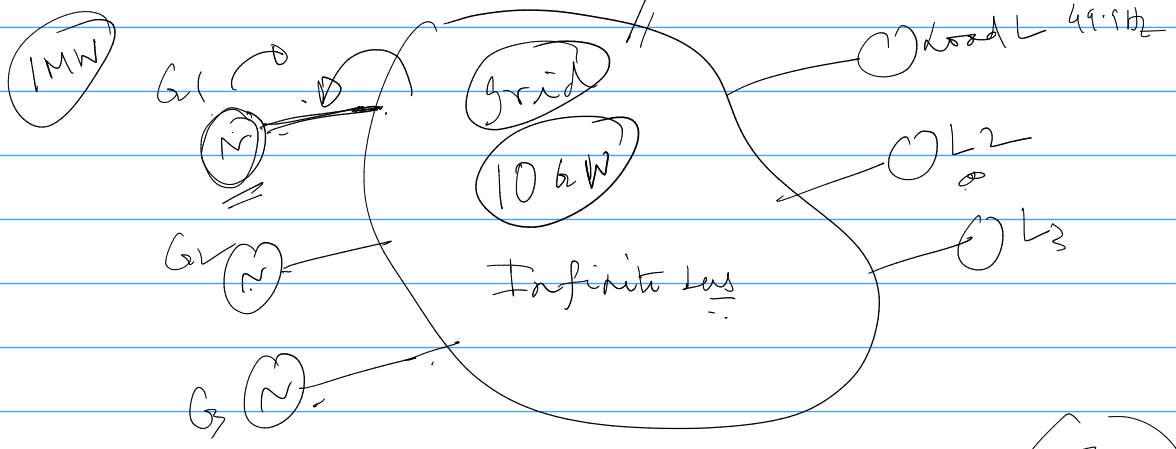
We can use single phase equivalent ckt



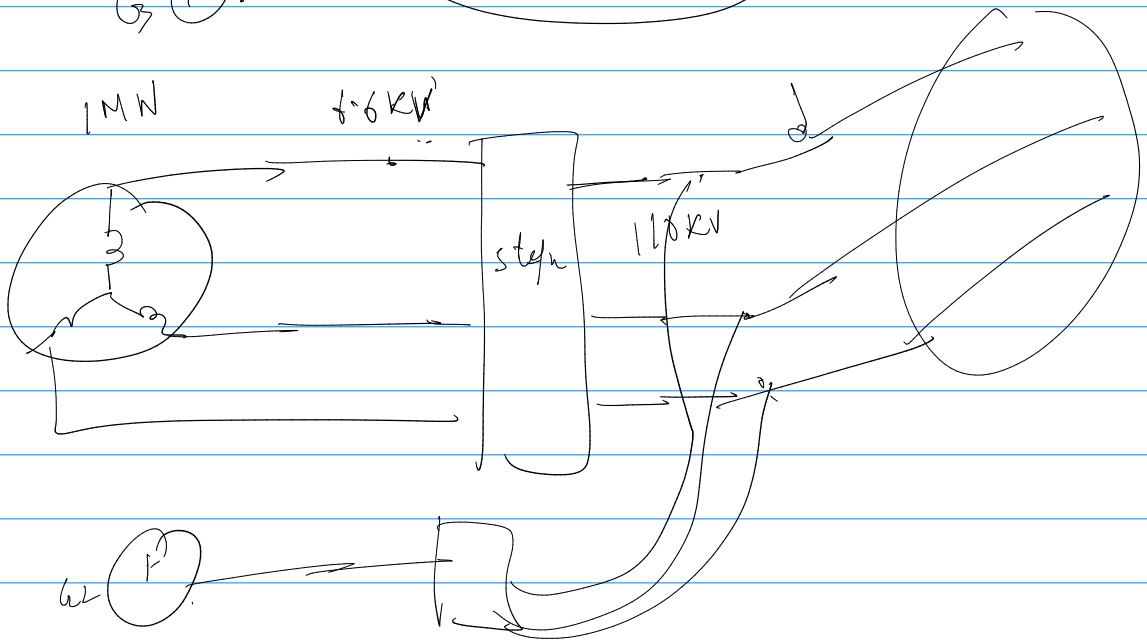


Synchronization and Infinite bus

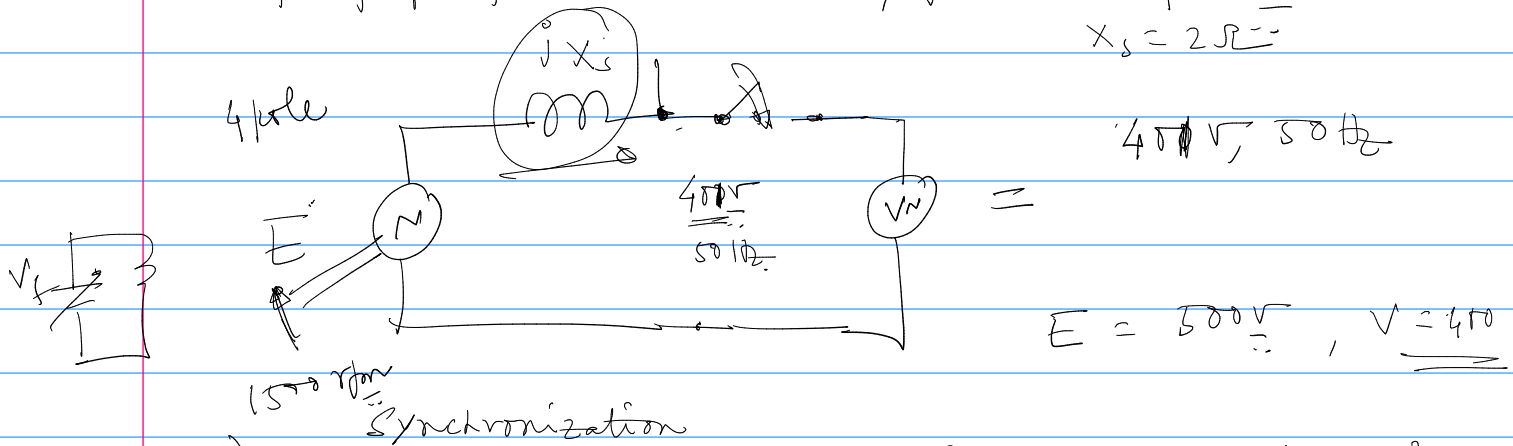




6 KV



Property of infinite bus → V, f remain fixed



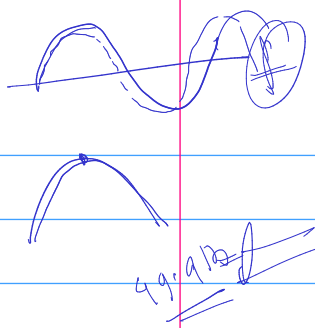
Synchronization

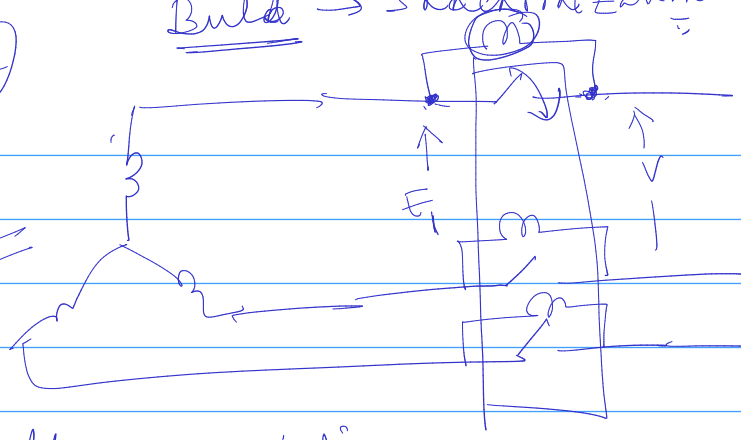
1) Terminal voltage should remain same as V (infinite bus)

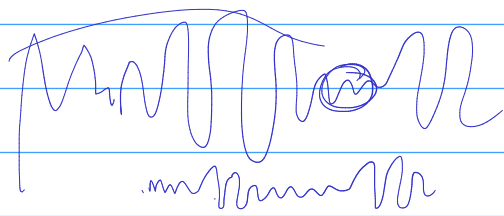
2) Frequency of Syn. g should be same as grid freq.

3) Keep the same phase sequence of the m/c and infinite bus.

Build → synchronization // synchroscope

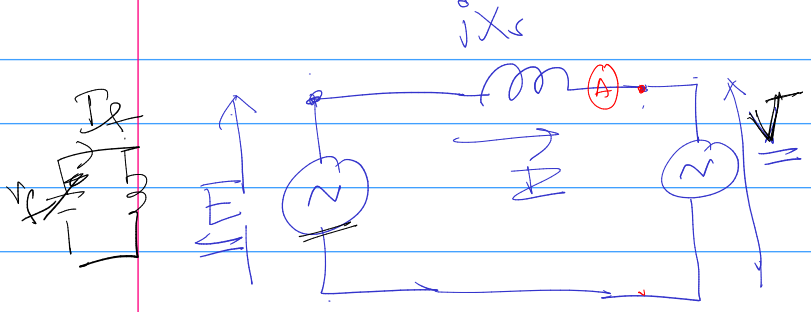




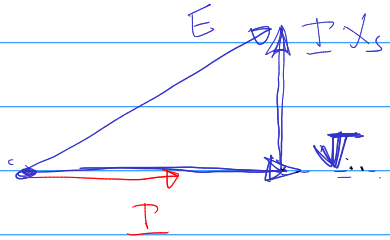


Voltage regulation

IX_s ⊥ V



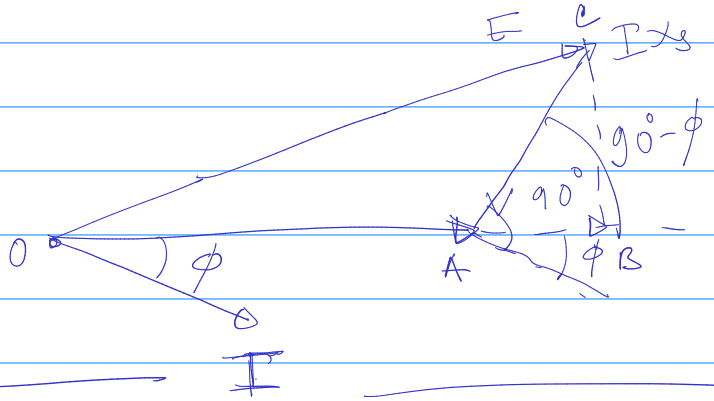
I = u.pf



$$E = V + jIX_s$$

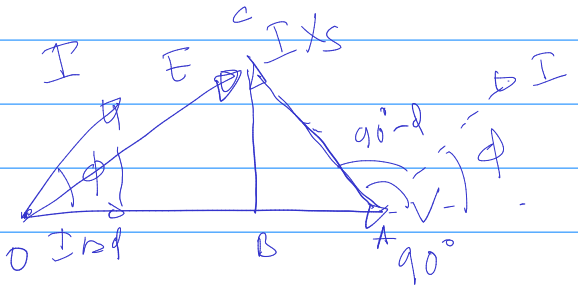
$$E = \sqrt{V^2 + (IX_s)^2}$$

Lagging p.f



$$E = \sqrt{(OA+AB)^2 + (BC)^2} = \sqrt{(V + IX_s \sin\phi)^2 + (IX_s \cos\phi)^2}$$

Leading p.f



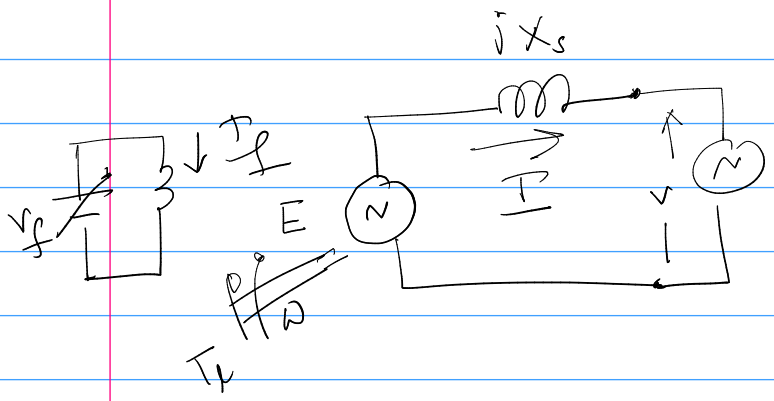
$$OC = E = \sqrt{(OB)^2 + (BC)^2} \qquad E = \sqrt{(V - IX_s \sin\phi)^2 + (IX_s \cos\phi)^2}$$

28th class

10th March 2024

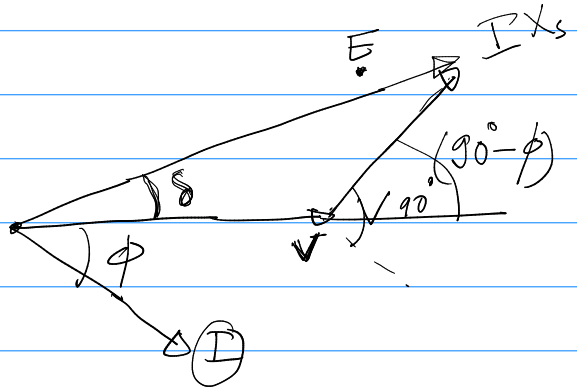
Voltage regulation = $\frac{E - V}{V} \times 100\ \%$

E = excitation voltage, V = terminal voltage



$I = \left(\frac{E - V}{jX_s}\right)$

I = lagging p.f.. ($\cos\phi$)



δ = load angle

(E & V)

Power delivered to the load = $(VI\cos\phi)$

$S = (VI^*)$

$P = Re(VI^*)$

$Q = Im(VI^*)$

$S = VI$

$= V \cdot I \angle \pm\phi$

$= VI\cos\phi - jVI\sin\phi$

$= P - jQ$

$S = P \pm jQ$

$P = Re[VI^*]$

$= Re\left[V \cdot \left(\frac{E}{X_s}\angle\delta - 90^\circ - \frac{V}{X_s}\angle -90^\circ\right)^*\right]$

$= Re\left[V \cdot \left(\frac{E}{X_s}\angle 90^\circ - \delta - \frac{V}{X_s}\angle 90^\circ\right)\right]$

$= Re\left[\frac{EV}{X_s}\cos(90^\circ - \delta)\right] - Re\left[\frac{V^2}{X_s}\angle 90^\circ\right]$

$= 0$

$\boxed{P = \frac{EV}{X_s}\sin\delta}$

$I = \frac{E - V}{jX_s}$

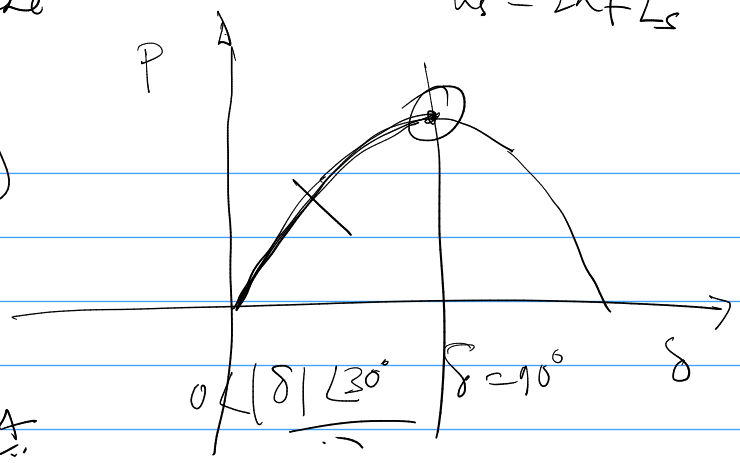
$I = \frac{E\angle\delta - V\angle 0}{X_s\angle 90^\circ}$

$= \frac{E}{X_s}\angle\delta - 90^\circ - \frac{V}{X_s}\angle -90^\circ$

As it is a 3-ϕ machine

$$P = \frac{3EV}{X_s}\sin\delta$$

$X_s = 2\pi f L_s$



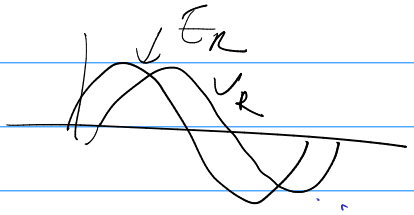
$\left|\frac{EV}{X_s}\right|_{max}$

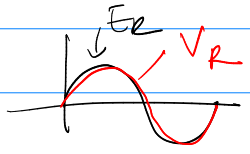
(I) synchronization

(1) $|E| = |V|$ (2) phase seq. should be same.

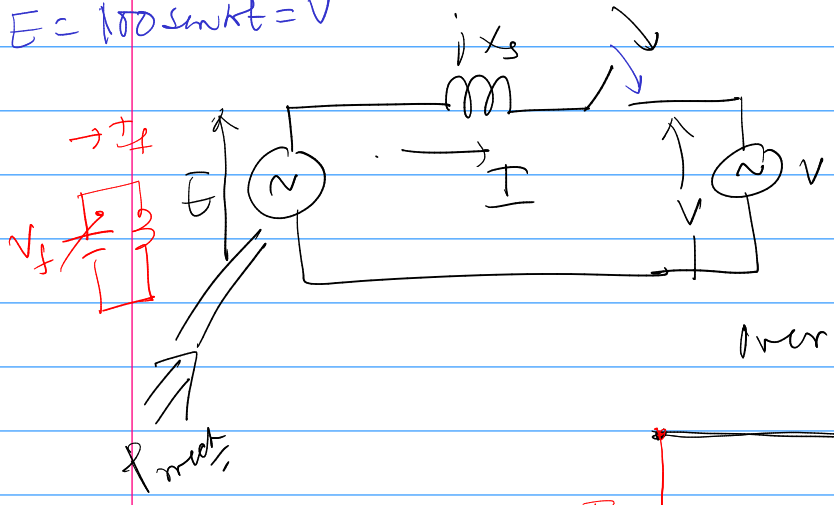
(3) phase difference between any two phase voltage is zero (E_R & V_R)

(4) same freq

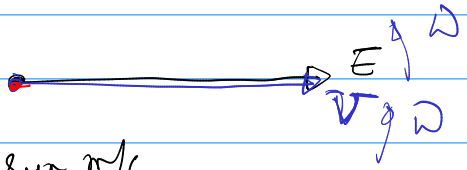




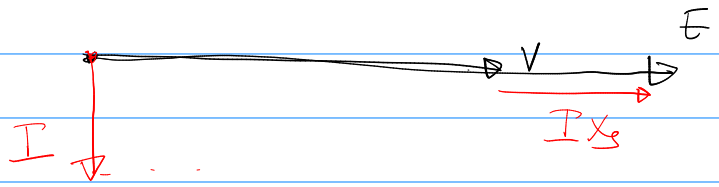
$E = 100\sin\omega t = V$



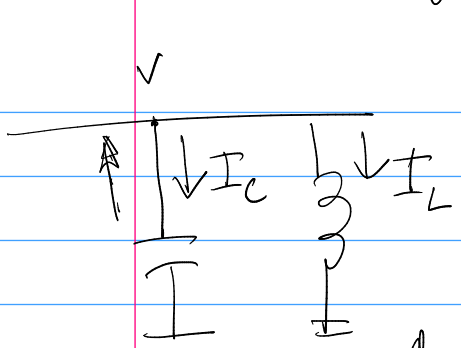
$E = V + jIX_s\angle 90°$

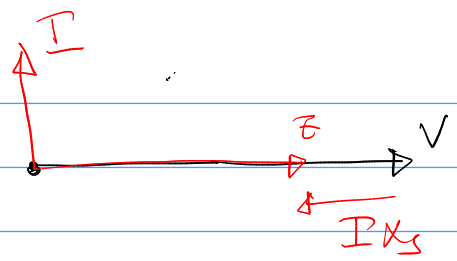


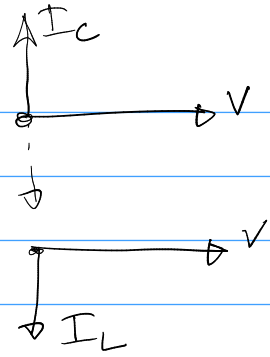
over excited syn m/c



Leading current (underexcite the machine)



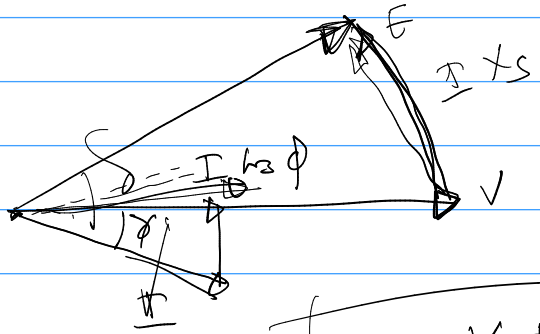




Overexcited syn m/c $\Rightarrow$ L, C ✗ ✓

Underexcited syn m/c $\Rightarrow$ L, C ✓ ✗

$$P = \frac{3EV}{X_s}\sin\delta$$



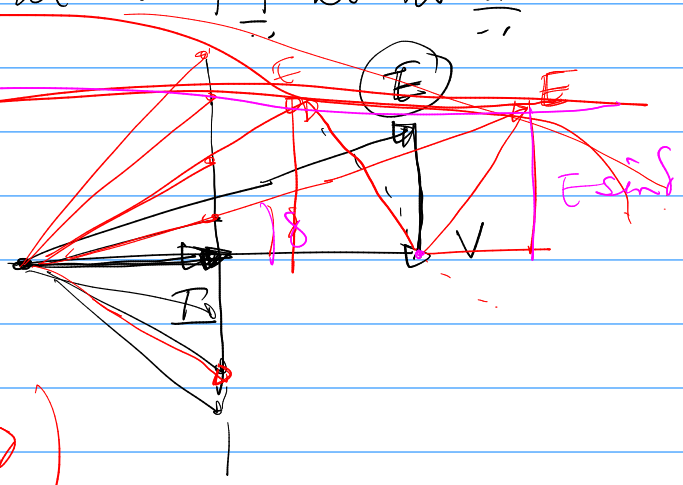
$$3VI\cos\phi = \frac{3EV}{X_s}\sin\delta$$

$$E = V + jIX_s$$



I would like to operate at u.p.f at load

$$E = V + jIX_s$$



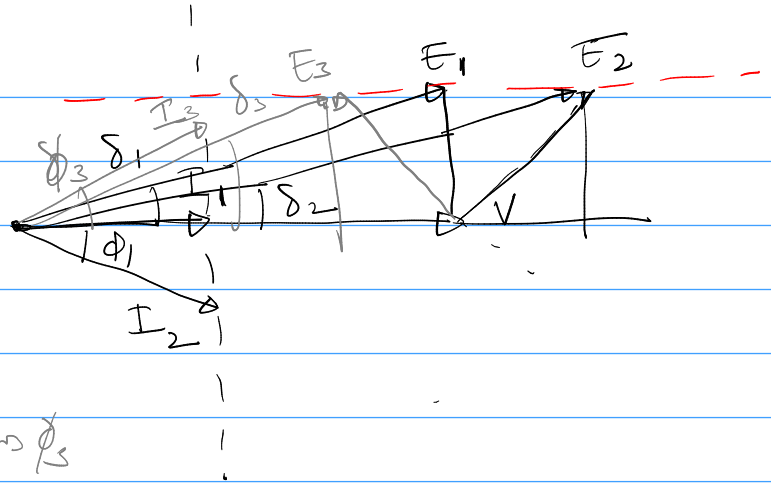
Real power is fixed

reactive current is fixed

$P = 3VI$ (E), (I)

$$3VI\cos\phi = \frac{3EV}{X_s}\sin\delta$$

$P = 3VI\cos\phi = \frac{3EV\sin\delta}{X_S}$



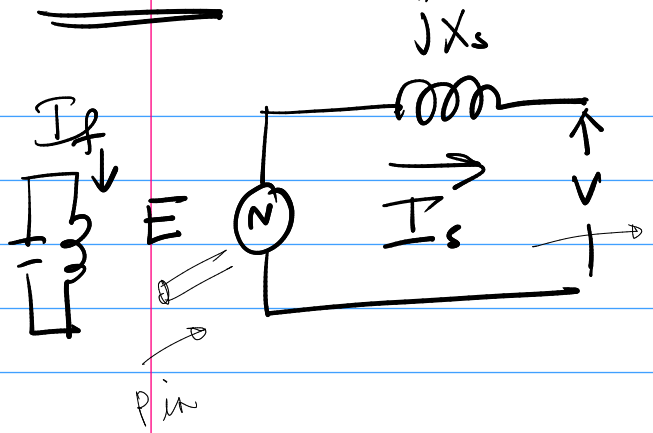
$E_1 \sin\delta_1 = E_2 \sin\delta_2$

$= E_3 \sin\delta_3$

$I_1 \cos\phi_1 = I_2 \cos\phi_2 = I_3 \cos\phi_3$

29th class

12/03/2026



$$I_s = \frac{E - V_s}{jX_s}$$

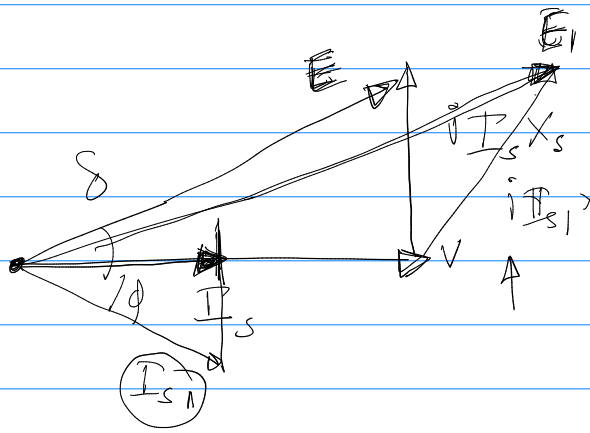
$$3VI_s \cos\phi$$

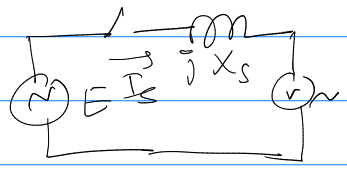
$$\frac{3EV}{X_s}\sin\delta$$

Unity p.f condition

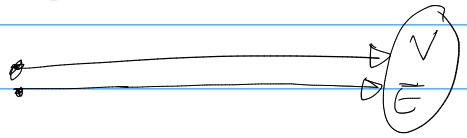
$I_s = I_{s1}\cos\phi$

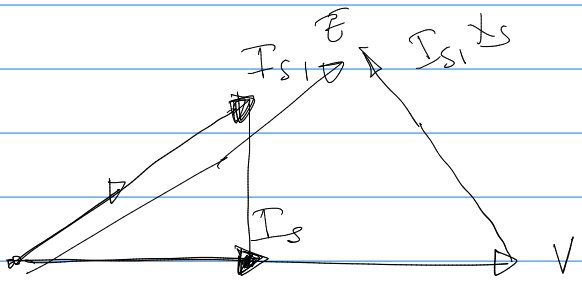
$|E| > |V|$





Overexcited/ underexcited



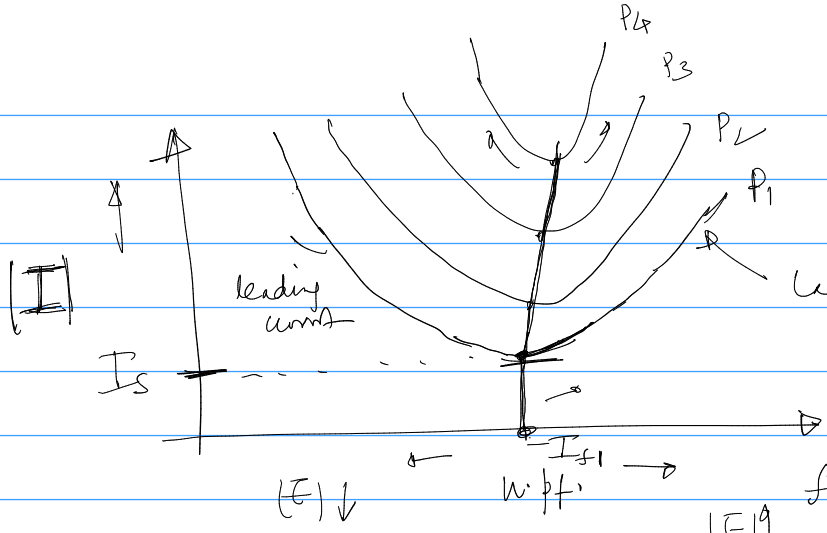


$3VI_s\cos\phi$

$|E| < V$

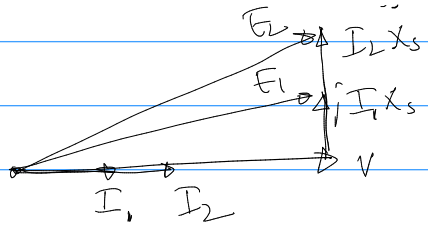
underexcited

$P_4 > P_3 > P_2 > P_1$



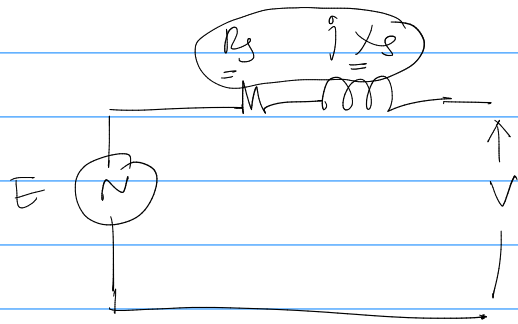
$P = 3VI\cos\phi$

V curve for a synchronous generator



$|E_2| > |E_1|$

$I_{f2} > I_{f1}$



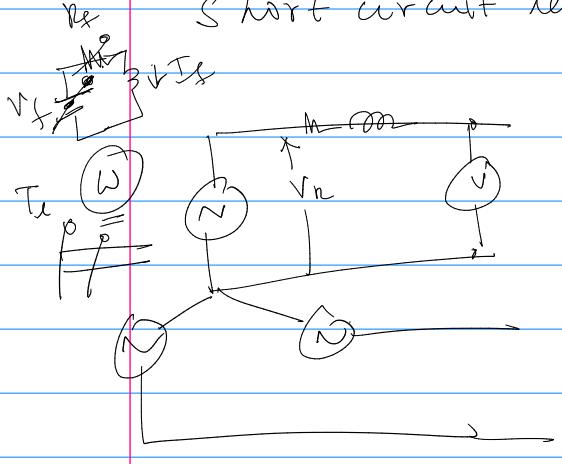
$R_s \simeq X_s$

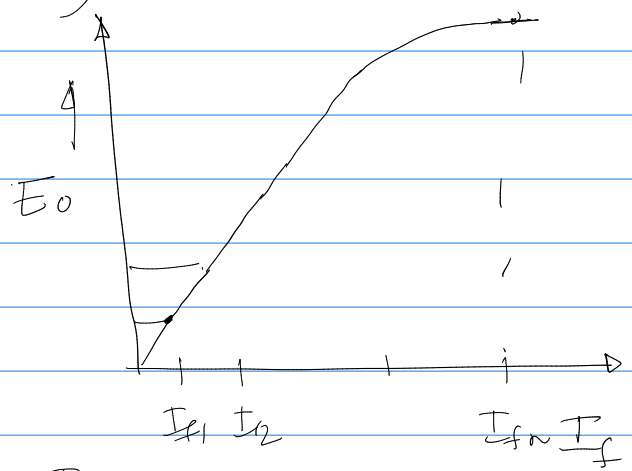
synchronous Impedance method

R_s, X_s

Open circuit test (O.C.C)

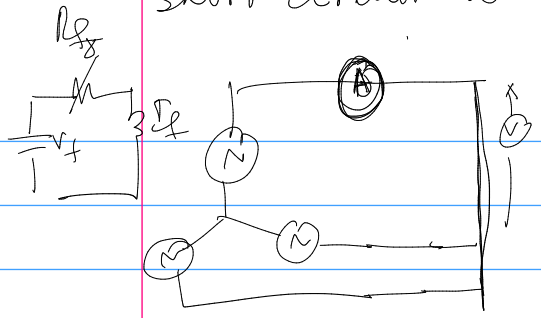
Short circuit test (S.C.C.)

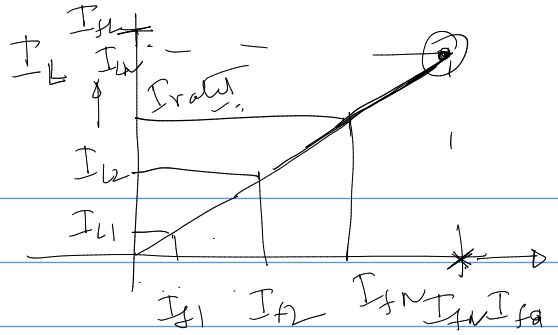


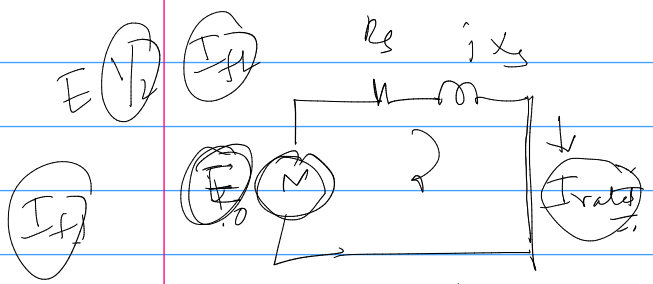


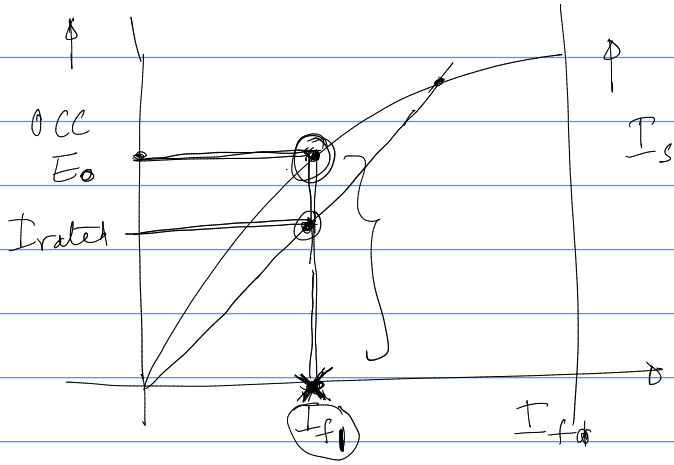
$E_0 \propto I_f$

Short circuit test



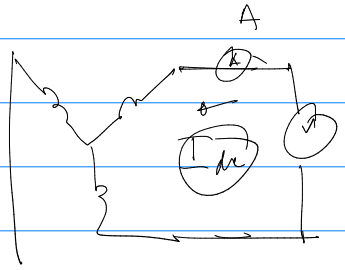


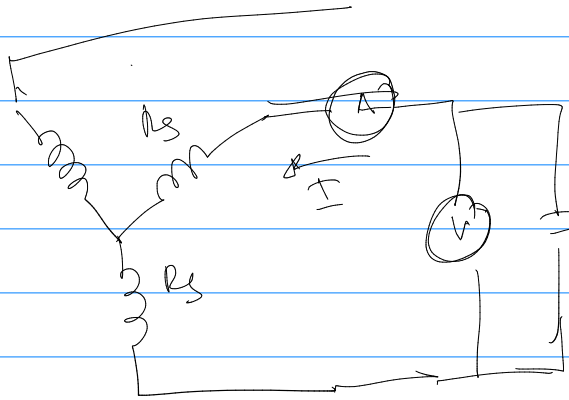




$$Z_s = E_0 / I_{rated}$$

$$X_s = \sqrt{Z_s^2 - R_s^2}$$





$$\frac{V}{I} = Z_s = 2R_s$$

Syn m/c $\Rightarrow$ 8 pole, 50 Hz $L_s = 5$ mH

(1) $N_s = ?$ $\quad f = \frac{PN}{120}$

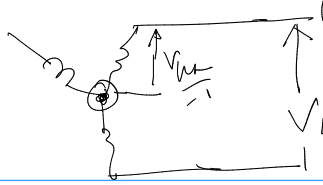
(2) $X_s = ?$ $(2\pi \times 50 \times 5)$ mH $\quad N_s = \frac{50 \times 120}{8} = 750$ rpm

$= 1.57\ \Omega$

Nameplate

440 V, 25 KVA, 50 Hz, Y connected Salient-sheet pole, 8 pole Syn m/c

Isorbm



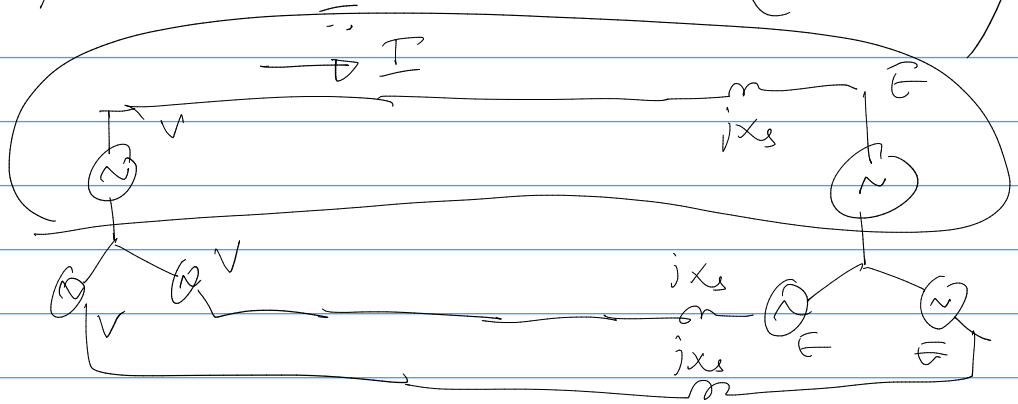
50Hz $V_{LL} = 400\,V\,(rms)$

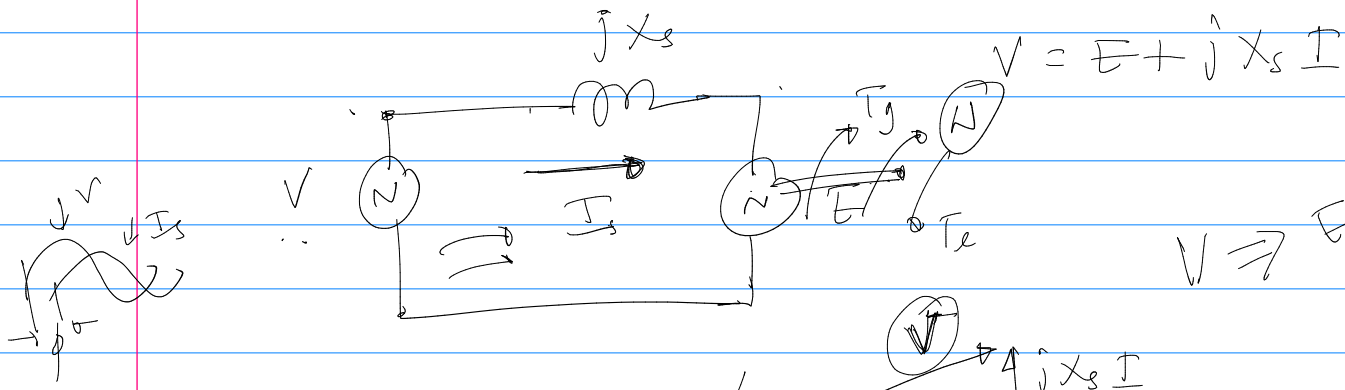
$V_{ph} = 400/\sqrt{3}$

$S = 3 V_{ph} I_{ph} = \sqrt{3}\, V_L I_L$

$I_L = \dfrac{25 \times 10^3}{\sqrt{3} \times 400}$

synchronous motor

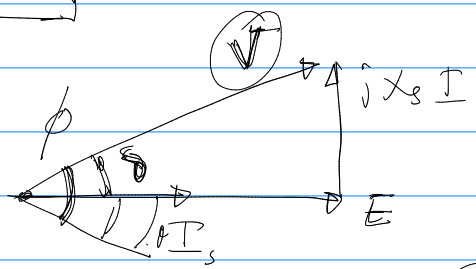




$V = E + jX_s I$

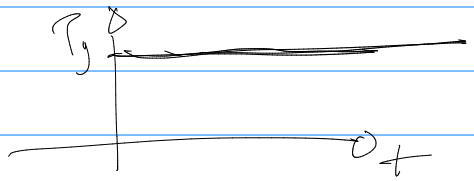
$V \Rightarrow E$

$|V| > |E|$



$3 V I_s \cos\phi = T_g \cdot \omega = 3 \dfrac{EV}{X_s} \sin\delta = 3 E I_s \cos\theta$

$T_g = \dfrac{3EV}{X_s \cdot \omega_s} \sin\delta$



$T_g = \dfrac{3EV}{\omega_s} \sin\delta$

Synchronous machine is used as a var compensator

